# Light Collection in a Polished Plastic Scintillator Read Out by Side-Mounted SiPMs: a Geant4 Matrix Simulation Study

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**Abstract**

We describe a Geant4 simulation of a 250 mm×250 mm×10 mm polished plastic scintillator (PVT, UPS-92S/EJ-200 class) read out by side-coupled silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. The detector is exposed to a vertical 4 GeV muon pencil beam. We present the geometry and the optical-physics implementation in detail—the Geant4 *unified* surface model, the finish/type combinations used to represent four wrapping configurations (bare/black holder, $TiO_2$ paint, aluminium foil, glossy paper), the photocathode efficiency model that drives photodetection, and the per-event optical accounting that feeds the analysis. A 48-file matrix scan (4 SiPM layouts × 4 reflectors × 3 beam positions, 1000 events each), complemented by a whole-plate irradiation (25 000 events per configuration and wrapper), yields collection efficiencies that reproduce only part of the experimentally observed ordering. The glossy-paper wrapper is the brightest and $TiO_2$ paint the weakest, with the specular wrappers the most position-uniform, but the bare plate is *not* the weakest collector: modelled with its true 1.2 mm air gap to the black holder, the polished plate is a near-lossless total-internal-reflection light guide, second in brightness only to glossy paper and nearly uniform over the whole plate ($\kappa_{2D} \approx 1.1$–$2.4$), while the diffuse $TiO_2$ paint is the most position-dependent. We report per-event photon yields, photoelectron spectra, wavelength distributions, position-resolved efficiency maps, and a loss study that quantifies the surface roughness—or plate–holder optical contact—that would be required to make the bare plate the weakest collector. Two implementation caveats are documented: a rare intermittent multi-threaded stall in the optical tracking, and the bare-plate ordering discrepancy above.

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## 1 Introduction and motivation

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers and veto systems for cosmic-ray muon tomography [18, 19]. The amount of scintillation light that reaches the photosensor is governed almost entirely by how the non-instrumented faces of the plate are treated: a polished plate in air traps light by total internal reflection (TIR) and can be a nearly lossless light guide, whereas a plate wrapped in an absorbing or diffusive material loses photons at every bounce. Understanding this trade-off quantitatively is the central design question, and it is precisely where simulation is most useful, because the photon transport is a multi-bounce random walk that is difficult to model analytically [7].

This report documents a Geant4 (v11.3.2) [2, 3] simulation built to address that question for the specific geometry and surface treatments of Rao *et al.* [1]. The deliverable is a self-contained "matrix scan": every combination of four SiPM layouts and four reflector treatments, at three beam positions on the plate. The goal is not to fit a single number but to test how far the simulation reproduces the *ordering* of light-collection efficiency that the experiment observes, and to expose the physics behind both the parts it reproduces and the part it does not.

The remainder of this document is organised as follows. Section 2 describes the detector model: the scintillator, the SiPM read-out, the four read-out layouts (including the wavelength-shifting fibre clamp of config 4), the reflector treatments, and the optical-physics implementation, all verified line-by-line against the Geant4 source. Section 3 presents the matrix simulation—its beam and patch setup, and the resulting collection efficiencies, photon/photoelectron spectra, and position-resolved maps. Section 4 presents the complementary whole-slab irradiation, which samples the full plate uniformly. Section 5 documents known caveats, and Sec. 6 summarises.

## 2 Detector model

### 2.1 Scintillator plate

The active volume is a rectangular PVT scintillator $250 \times 250 \times 10\,\text{mm}$, mirroring the UPS-92S plate of Ref. [1]. It is assigned the Geant4 NIST base material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical properties table built from the EJ-200 and UPS-923A data sheets [11, 12, 13]. The optical constants—the refractive index ($n = 1.58$, essentially flat across the visible), light yield, emission peak, bulk attenuation length, scintillation decay and rise times, and the

Birks constant [14, 15]—are collected in the consolidated parameter table of Sec. 2.5 and are not repeated here. Two entries are deliberate modelling choices rather than quoted numbers: the bulk attenuation length is peaked at 3.8 m but falls steeply in the UV to model wavelength-shifter self-absorption, and the emission spectrum (Fig. 3) is essentially zero below 380 nm and above 550 nm. Both scintillation and Čerenkov photons are produced and tracked (Sec. 2.7); the optical grid spans 310 nm to 700 nm so that the Čerenkov continuum, which extends well below the scintillation band, is covered.

### 2.2 SiPM readout and coupling

Each SiPM [8] is modelled mechanically as a $4 \times 4 \times 0.6$ mm box of "SiPM epoxy" ($n = 1.55$), coupled to a side face of the plate through a 0.1 mm silicone grease pad ($n = 1.46$). The coupling chain is therefore *scintillator* → *grease* → *SiPM*, with Fresnel transmission at each dielectric interface and photodetection at the final grease/SiPM boundary.

Four layouts are studied (schematic in Fig. 1):

**Config 1 (8 SiPMs)** two SiPMs per side face, at ±50% of the half-length.

**Config 2 (4 SiPMs)** one SiPM per side face, at the *same* ±50% offsets as config 1—i.e. config 1 with one device removed per side, the removed slot rotated from side to side so the layout remains four-fold symmetric.

**Config 3 (2 SiPMs)** both SiPMs on the *same* face—the top ($+y$) edge—at ±50%, matching T1/T2 of Ref. [1].

**Config 4 (1 SiPM + WLS clamp)** a single SiPM reading out a wavelength-shifting (WLS) fibre clamp along the $-x$ narrow face (Sec. 2.4).

SiPM footprints are clamped so they cannot overhang the slab face.

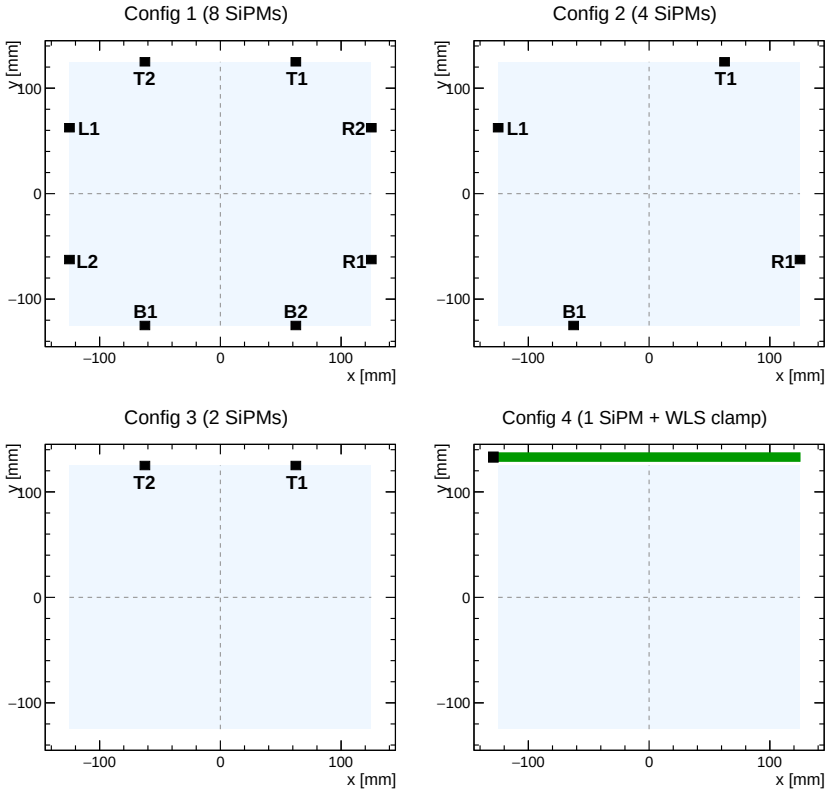



**Figure 1:** Schematic plan view of the four detector layouts on the 250 mm×250 mm plate footprint. Black squares mark the side-mounted SiPMs (drawn oversized for visibility; the active area is 4 mm×4 mm), and the dashed grey lines are the $x = 0$, $y = 0$ axes. Config 1 places two SiPMs on each of the four edges; config 2 places one SiPM per edge at the same offsets as config 1 (one device removed per side); config 3 places both SiPMs on the top ($+y$) edge; config 4 uses a single SiPM reading out a WLS fibre clamp, drawn along the top ($+y$) edge with the black read-out SiPM at its left end (green bar), following Fig. 3(b) of Ref. [1]. The inwards-offset labels name the eight side-mounted sensors (T/B/R/L = top/bottom/right/left); on each edge "1" marks the sensor nearer the corner reached when travelling clockwise around the perimeter, so "1" sits at the top-right (T1), bottom-right (R1), bottom-left (B1) and top-left (L1) corners. This is the scheme used in the light-sharing discussion below. The same black-square markers are overlaid on the collection-efficiency maps (Figs. 14–17).

### 2.3 Reflector treatments

Four treatments are applied as a *logical skin* over the non-instrumented surfaces of the plate; the grease/SiPM patches override the skin locally with dedicated *border* surfaces. The mapping from each physical treatment to the Geant4 optical-surface parameters is the heart of the model—it determines the light-collection ordering—so it is given in full. Table 1 lists every property that is actually attached to each surface, and the subsubsections below describe each reflector physically, state how it is modelled, and assess how faithfully the model represents the real object. (A fifth finish, PTFE tape, exists in the code for completeness but is not part of the reference study.) The strong sensitivity of light collection to the wrapping and reflector choice is well established experimentally [21].

**Table 1:** Geant4 optical-surface parameters for the four reflectors, exactly as attached in `src/GeometryMuonScint.cc`. $R(\lambda)$ is tabulated on the shared ten-point 360 nm to 650 nm grid and quoted here as its range; the full arrays are in the code. A dash marks a property that is neither set nor used for that finish.

| | **Bare** (black holder) | **$TiO_2$** (EJ-510 paint) | **Al foil** (+ grease) | **Glossy** (photo paper) |
|---|---|---|---|---|
| Physical contact | 1.2 mm air gap | direct (optical) | grease film | air gap |
| Geant4 surface object | none* | skin | skin | skin |
| Type | `dielectric_dielectric` | `dielectric_dielectric` | `dielectric_dielectric` | `dielectric_dielectric` |
| Model | `glisur`* | `unified` | `unified` | `unified` |
| Finish | `polished`* | `groundfront-painted` | `polished-backpainted` | `polished-backpainted` |
| `sigma_alpha` | — | 0.10 rad | 0 | 0 |
| `REFLECTIVITY` $R(\lambda)$ | — | 0.955–0.983 | 0.904–0.925 | 0.900–0.955 |
| Backing `RINDEX` | — | — | 1.46 (grease) | 1.0003 (air) |
| `SPECULARSPIKE` $f_{ss}$ | — | 0 | 1 | 1 |
| `SPECULARLOBE` $f_{sl}$ | — | 0 | 0 | 0 |
| `BACKSCATTER` $f_{bs}$ | — | 0 | 0 | 0 |
| Total internal reflection | yes | no | yes | yes |
| | (Fresnel/TIR) | (Lambertian) | ($\leq 67.5°$) | (full cone) |

*The bare plate carries *no* surface object: Geant4 then applies its built-in `glisur/polished dielectric_dielectric` default, with Fresnel and TIR taken from the material refractive indices.

### 2.3.1 Bare plate in its black holder

*Physical object.* The polished UPS-92S plate is clamped into a black 3D-printed ABS enclosure, its faces simply resting near the wall.

*Model.* No wrapping skin is attached. The plate keeps its native polished scintillator ($n = 1.58$) against air ($n = 1.0003$) across a real 1.2 mm gap; light below the critical angle ($\arcsin(1/1.58) = 39.3°$) refracts across the gap and is absorbed in the holder, whose black ABS has an attenuation length of 1 µm, while light above it is returned by total internal reflection. Fresnel reflection and TIR are taken from the material `RINDEX` tables.

*Fidelity.* This is the least faithful configuration and the source of the report's central discrepancy. The model assumes the plate is optically *decoupled* from the holder everywhere. A real clamped plate touches the holder at scattered high spots, and each contact forms a frustrated-TIR sink that removes trapped light; surface scratches do the same. Those loss channels are absent here, so the modelled plate acts as a near-lossless light guide and is one of the *brightest* collectors, whereas the reference measurement gives it by far the *lowest* signal. We keep the clean air-gap model because it is the defensible reading of the stated construction (a plate set in a holder), and quantify in Sec. 5 the roughness or absorption that would be needed to reproduce the measurement.

### 2.3.2 Titanium-dioxide paint (EJ-510)

*Physical object.* A white, diffuse-reflectance paint applied directly to the plate (three coats, $\approx 0.11$ mm), in optical contact with the scintillator.

*Model.* `groundfrontpainted` with a purely Lambertian reflection: a random draw against `REFLECTIVITY` returns the photon with a cosine-law distribution and *no* specular or TIR component ($f_{ss} = f_{sl} = f_{bs} = 0$). The reflectance rises from 0.955 in the UV to a plateau of 0.983 near 470 nm.

*Fidelity.* The Lambertian/no-TIR treatment is the physically correct description of light emerging from a thick, strongly scattering paint, and is this wrapper's strongest feature. The *absolute* reflectance curve is its weakest: no public spectral-reflectance data exist for EJ-510, so

$R(\lambda)$ is a fitted representative white-paint curve ($\approx 0.96$–$0.98$), not a measurement of the actual coating, and no angular dependence of the diffuse lobe is modelled. The `sigma_alpha` value is set for bookkeeping but is inert at this finish (`groundfrontpainted` reflects diffusely regardless of the facet normal).

#### 2.3.3 Aluminium foil

*Physical object.* Commercial shiny aluminium foil wrapped around the plate; per Ref. [1] optical grease is applied in places so that the foil contacts the scintillator rather than leaving an air gap.

*Model.* `polishedbackpainted` with the backing refractive index set to the coupling grease ($n = 1.46$). The scintillator/grease interface is a normal Fresnel/TIR boundary, but since the index step is small the critical angle is wide ($\arcsin(1.46/1.58) = 67.5°$): most light refracts into the grease film and is returned by the foil as a pure *specular* reflection ($f_{ss} = 1$). `REFLECTIVITY` is the normal-incidence reflectance of evaporated aluminium computed from the McPeak *et al.* $n, k$ data [20] (0.904 at 650 nm rising to 0.925 at 360 nm).

*Fidelity.* Two idealisations. (i) The tabulated reflectance is that of a smooth evaporated film and is therefore an *upper bound*: household foil is rougher and carries a native oxide, both of which lower $R$ and add a small diffuse component we do not model. (ii) The foil is assumed to contact the plate uniformly through an ideal grease layer, whereas real wrapping leaves wrinkles and trapped air. The grease-contact assumption itself follows the reference construction and is the reason this wrapper outperforms glossy paper in the model.

#### 2.3.4 High-gloss photo paper

*Physical object.* A commercial super-high-gloss photographic paper (Print4Life, $300\,\mathrm{g\,m^{-2}}$) wrapped around the plate with an air gap (no grease).

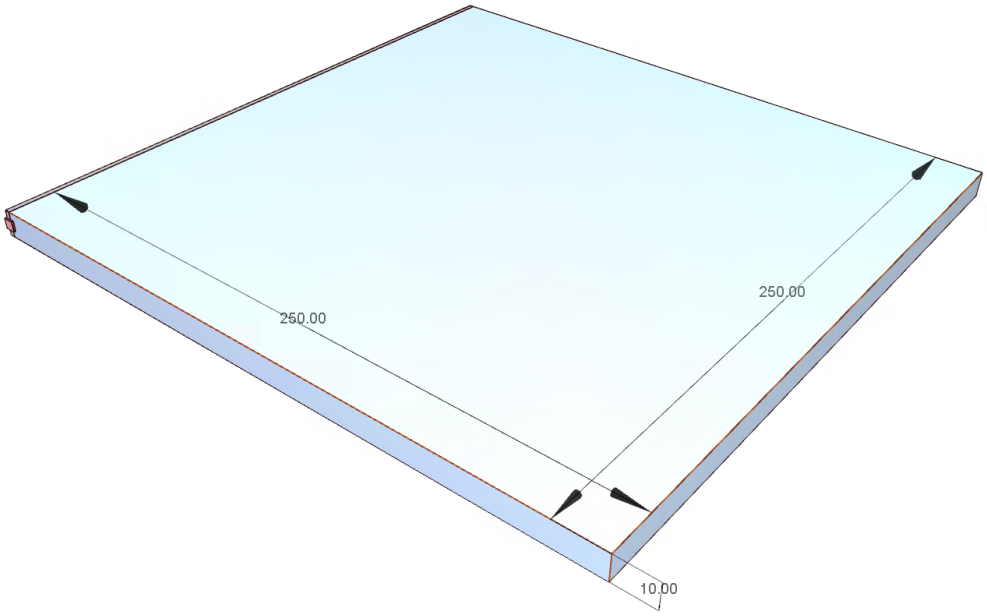
*Model.* `polishedbackpainted` with the backing index set to air ($n = 1.0003$), so the plate retains its full TIR cone and only the sub-critical fraction crosses the gap to the paper, which returns it as a specular reflection ($f_{ss} = 1$) with a fitted $R(\lambda) \approx 0.900$–$0.955$—slightly lossier than the aluminium.

*Fidelity.* The reference reports glossy paper and aluminium foil as the two best wrappers (large, nearly equal signals), which motivates modelling glossy paper as a *slightly lossy mirror* rather than a diffuser. This is a model choice rather than a measurement: real photo paper is not a true mirror and has a diffuse component, and its $R(\lambda)$ is again fitted (no measured data), so this is the reflector whose optical model is least constrained by independent data.

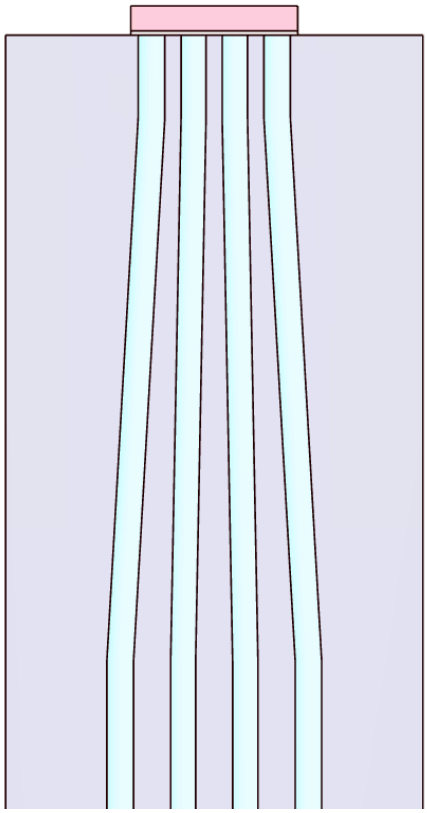
### 2.4 Config 4: wavelength-shifting fibre clamp

The fourth layout follows the single-SiPM, WLS-readout variant of Ref. [1]. Four 1.0 mm-diameter Kuraray Y-11(200) wavelength-shifting fibres [16, 17] run the full 250 mm length of the $-x$ narrow face in a single row. Over most of the length they are parallel at a 1.5 mm centre-to-centre pitch—so a solid web of clamp material separates the milled 1.1 mm channels—and converge over the last 15 mm to a touching 1.0 mm bundle that presents all four ends to a single 4 mm×4 mm SiPM at the $-y$ end; their far $+y$ ends are closed by individual aluminium mirror caps. A thin 0.10 mm grease film couples the plate face to the clamp, and each fibre protrudes 0.05 mm through its channel mouth into that film, keeping the grease continuous against the scintillator. The deliberate 0.05 mm air annulus between the 1.0 mm fibre and its 1.1 mm channel is essential: the modelled cladding ($n = 1.49$) exceeds the grease ($n = 1.46$) only marginally, so a snug fit against the $n \approx 1.55$ clamp would frustrate core–cladding total internal reflection. The outboard face of the black clamp carries a mirror skin. Because the core–cladding index step is small, guiding relies on the cladding staying optically isolated from the clamp, which the air annulus guarantees. The WLS process is enabled automatically when this geometry is built. In the

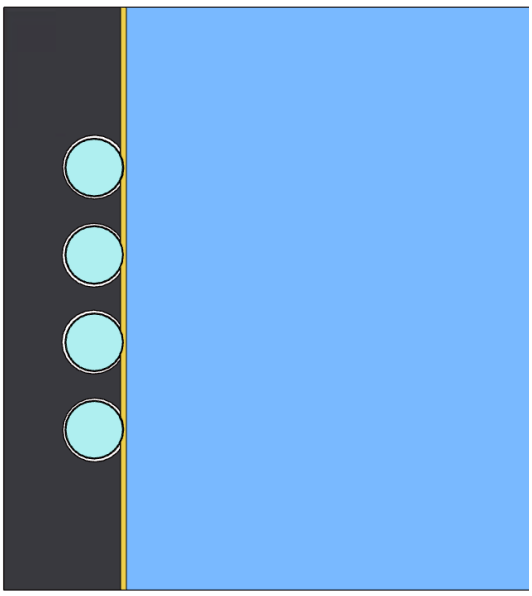
report's top-down frame the clamp runs along the *top* edge with the single read-out SiPM at the *top-left* corner, as in Fig. 3(b) of Ref. [1].




**(a)** Three-dimensional view of the complete config-4 assembly, with the principal dimensions annotated.



**(b)** Top view of the black fibre clamp: the four fibre channels and the read-out SiPM at the end of the converged bundle (report-frame top-left).



**(c)** Cross-section through the clamp: the scintillator, the thin grease film in the slab–clamp gap, the four fibre discs seated in their 1.1 mm channels with a 0.05 mm air annulus, and the outboard black clamp.

**Figure 2:** Geometry-only views of the config-4 WLS-fibre clamp, rendered from the Geant4 VRML geometry export (*no* beam-on). Following the event-display colour scheme, fibre cores are pale green, cladding dark green, the grease film amber, the black clamp and holder dark grey, and the read-out SiPM red. The four 1.0 mm fibres sit on the $-x$ narrow face in 1.1 mm channels separated by clamp webs, with only a thin grease film in the slab–clamp gap; they converge to a bundle read out at $-y$ by one SiPM and mirrored at $+y$.

### 2.5 Summary of detector parameters

The principal parameters of the three active components—the scintillator plate, the WLS fibre, and the SiPM—are collected in Table 2, alongside the corresponding values quoted for the reference experiment of Ref. [1]. Where a quantity is not reported in Ref. [1] or is not modelled here, a dash is shown; the simulation follows the experimental values wherever they are available, and the optical model of Sec. 2.7 is built entirely from the values in the last column.

**Table 2:** Principal parameters of the scintillator, the WLS fibre, and the SiPM, comparing the reference experiment of Ref. [1] with the values adopted in this simulation.

| Parameter | Ref. [1] | This simulation |
|---|---|---|
| *Scintillator (UPS-92S / PVT)* | | |
| Material | UPS-92S (AdvTECH UK) | PVT (`G4_PLASTIC_SC_VINYLTOLUENE`) |
| Dimensions | $250 \times 250 \times 10$ mm | $250 \times 250 \times 10$ mm |
| Refractive index | 1.58 | 1.58 |
| Density | 1.05 g cm$^{-3}$ | 1.032 g cm$^{-3}$ |
| Light output | 50–60% of anthracene ($\approx 8$ to $10 \times 10^3$ ph/MeV) | $1.0 \times 10^4$ MeV$^{-1}$ |
| Emission peak | 425 nm | 425 nm (2.917 eV) |
| Bulk attenuation length | 2000 mm | 3.8 m (peak, falling in the UV) |
| Scintillation decay time | 2.4 ns | 2.1 ns |
| Rise time | — | 0.9 ns |
| Birks constant | — | 0.126 mm MeV$^{-1}$ |
| Surface finish | all sides polished | all sides polished |
| *WLS fibre (Kuraray Y-11(200))* | | |
| Fibre diameter | 1.0 mm | 1.0 mm (core+clad) |
| Cladding | multi-cladding | single PMMA clad ($n = 1.49$) |
| Core refractive index | — | 1.59 |
| Absorption peak | 430 nm | 430 nm to 460 nm (WLS onset) |
| Emission peak | 476 nm | 476 nm |
| Attenuation length | $> 3500$ mm | $\gtrsim 5$ m (transport) |
| WLS decay time | — | 7 ns |
| Read-out arrangement | single row, 250 mm long | 4 fibres, 1.5 mm pitch $\rightarrow$ 1.0 mm bundle |
| Channel diameter | — | 1.1 mm (0.05 mm air annulus) |
| *SiPM (CEFLA S.C. ASD-NUV4S-P-40)* | | |
| Active area | $4 \times 4$ mm$^2$ | $4 \times 4$ mm$^2$ |
| Number of cells | 9340 | — |
| Peak sensitivity | 420 nm | 420 nm to 425 nm |
| Photon detection efficiency | 43% | 0.43 (peak) |
| Epoxy refractive index | 1.5115 | 1.55 |
| Optical coupling | BC-630 grease | 0.1 mm grease film ($n = 1.46$) |
| Gain / dark-count rate / capacitance | $10^6$ / 50–100 kHz/mm$^2$ / 0.84 nF | — (light transport only) |

### 2.6 Optical property tables

Figure 3 plots the three sets of optical constants actually fed to Geant4, over the same 310 nm to 700 nm range used in the transport. The left panel compares the normalised emission spectra: the scintillator peaks at 425 nm, while the Y-11 fibre re-emits in the green, peaked near 480 nm—the shift that makes the config-4 read-out spectrum green. The middle panel gives the absorption (attenuation) lengths on a logarithmic scale: the scintillator is transparent across the visible but its bulk attenuation collapses in the UV (the wavelength-shifter self-absorption band), whereas the fibre is effectively transparent red-ward of 500 nm and absorbs strongly below 480 nm (the WLS onset, marked by the dashed line). The right panel is the SiPM photocathode efficiency = PDE($\lambda$), peaking at 0.43 near 420 nm. These curves are exactly the tabulated points of `src/GeometryMuonScint.cc`; Geant4 interpolates them linearly, so the drawn segments are

what the simulation uses.

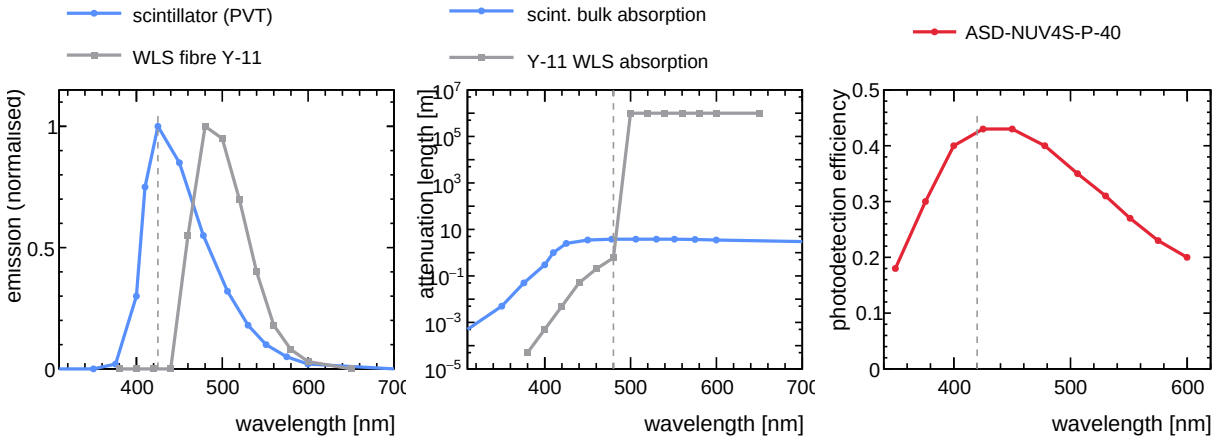



**Figure 3:** Optical property tables used by the simulation. *Left:* normalised emission spectra of the PVT scintillator (peaking at 425 nm) and of the Y-11 WLS fibre (peaking at 480 nm). *Middle:* absorption (attenuation) lengths on a logarithmic scale—the scintillator bulk absorption and the fibre WLS absorption. *Right:* SiPM photocathode PDE($\lambda$), peaking at 420 nm. The vertical dashed line in each panel marks the characteristic wavelength of that panel's process—scintillator emission peak at 425 nm (left), WLS absorption onset at 480 nm (middle), and SiPM PDE peak at 420 nm (right); the left and right lines therefore nearly coincide, whereas the middle line is red-shifted to the WLS band. Markers are the tabulated points; segments are the linear interpolation Geant4 applies.

### 2.7 Optical physics implementation

Optical photons are transported by the standard Geant4 optical physics module [4], with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes all active. Multiple scattering (Mie) is disabled; wavelength shifting (the `G4OpWLS` process) is enabled only for the config-4 fibre layout (Sec. 2.4) and inert for all others. This subsection describes the boundary physics and the photodetection model, and states explicitly the Geant4 behaviour that each modelling choice relies on, verified against the Geant4 source in `.refs/source/processes/optical`.

### 2.8 The boundary process and the unified surface model

Photon crossing of a dielectric interface is handled by `G4OpBoundaryProcess`, which selects an `G4OpticalSurface` from the unified surface model of Levin and Moisan [5] (whose key parameters for Geant4/GATE photon transport were later characterised [6]), using (*i*) a `G4LogicalBorderSurface` between two physical volumes or (*ii*) a `G4LogicalSkinSurface` on a logical volume. Reading the source (lines 319–347), the lookup is: a border surface between the pre- and post-step volumes is consulted first; if none exists, the skin surface of the daughter (or, failing that, the mother) logical volume is used. This establishes two facts on which the geometry relies:

1. a `G4LogicalBorderSurface` takes precedence over a `G4LogicalSkinSurface` at the same boundary—hence the grease and SiPM patches punch "holes" in the wrapping skin;
2. if no surface object exists at a boundary, the process falls back to the default lossless TIR behaviour (via `DielectricDielectric()`).

The boundary process requires a valid `RINDEX` on both media; if the property is absent it sets status `NoRINDEX` and kills the photon (lines 289–306, 436–453). Every material in the model therefore carries an explicit `RINDEX` table.

For the *unified* model the probability weights for the reflection sub-models are read from the surface MPT—`SPECULARSPIKECONSTANT` ($f_{ss}$), `SPECULARLOBECONSTANT` ($f_{sl}$) and `BACKSCATTERCONSTANT` ($f_{bs}$)—together with `REFLECTIVITY` (line 381) and, where applicable,

TRANSMITTANCE (line 394) and EFFICIENCY (line 390). The surviving probability $1 - f_{ss} - f_{sl} - f_{bs}$ is shared between diffuse (Lambertian) and bare reflection.

### 2.9 Reflector finishes

The four treatments use three distinct code paths in G4OpBoundaryProcess, which we verified:

- groundfrontpainted ($TiO_2$). The finish is handled by the "painted-front" branch (lines 462–488): a uniform random draw against REFLECTIVITY decides absorbed / transmitted / reflected, and reflection is set to a pure Lambertian distribution (LambertianReflection, line 481). There is *no* TIR at this finish—photons are either diffusely reflected or absorbed. This is the physically correct description of sub-surface scattering in a thick $TiO_2$ paint.
- polishedbackpainted (Al, glossy paper). This finish is routed to DielectricDielectric() (line 456–459). The scintillator/backing interface still exhibits Fresnel reflection and TIR; sub-critical photons cross it and are returned by the backing "paint" with reflectivity $R(\lambda)$ and a reflection distribution set by $f_{ss}, f_{sl}, f_{bs}$. We choose $f_{ss} = 1$ (pure specular spike), since both wrappers are smooth mirrors, and set the backing refractive index to distinguish the two physical situations. For the aluminium foil the backing index is the coupling grease ($n = 1.46$), following Ref. [1], which applies grease so the foil optically contacts the plate; the critical angle is then only $\arcsin(1.46/1.58) = 67.5°$, so most light refracts out of the plate and is returned specularly by the foil, whose reflectivity is the tabulated normal-incidence reflectance of evaporated aluminium. For glossy paper the backing index is that of air ($n = 1.0003$), so the plate keeps its full TIR cone and only the sub-critical fraction reaches the paper, which acts as a slightly lossier mirror.
- **Bare plate**. This is deliberately *not* given a wrapping skin. The plate keeps its plain polished boundary across a real 1.2 mm air gap to the holder shell; refraction across the gap plus absorption in the black holder bulk removes the sub-critical fraction, while super-critical light is retained by TIR. There is *no* absorbing skin on the plate itself.

### 2.10 Wavelength shifting

In the config-4 layout, photons that reach the fibre cores are absorbed and re-emitted isotropically at longer wavelength by the G4OpWLS process, which is configured through the fibre material's WLSABSLENGTH, WLSCOMPONENT and WLSTIMECONSTANT properties (Y-11 absorption peaks near 430 nm, re-emission near 490 nm). The process is switched on by the physics list when the WLS geometry is detected (the Geant4 optical-physics constructor reads the process activation at build time, so the flag is set before the base construction). Because the shifted emission no longer overlaps the fibre absorption band, the re-emitted light is transported losslessly along the fibre by TIR to the read-out SiPM—which is why the detected config-4 spectrum is green-shifted (mean $\sim$485 nm) relative to the direct-coupled configurations.

### 2.11 Photodetection: the SiPM photocathode

Photodetection is delegated entirely to Geant4, rather than being drawn manually in the sensitive detector. The grease/SiPM boundary is a dielectric_metal surface whose MPT carries the SiPM EFFICIENCY as a function of photon energy (equivalently, of the wavelength $\lambda$), i.e. the photon detection efficiency (PDE) of an ASD-NUV4S-P-40 device [10], peaking at $\approx 0.43$ near 420 nm, with REFLECTIVITY $= 0$.

The mechanism is worth stating precisely because it is the reason the sensitive detector needs no random draw of its own. In DielectricMetal() (lines 724–817), when the Fresnel–PDE roll places a photon in the "absorbed" channel, DoAbsorption() is called. Inside it (header, lines 305–320):

$$f_{\text{status}} = \begin{cases} \text{Absorption}, & \xi \geq \text{EFFICIENCY}(\lambda), \\ \text{Detection}, & \xi < \text{EFFICIENCY}(\lambda), \end{cases} \qquad \xi \sim U(0,1), \tag{1}$$

where EFFICIENCY($\lambda$) is the tabulated PDE. The absorbed photon is assigned a non-zero `LocalEnergyDeposit` *only* when the PDE draw succeeds. Back in `PostStepDoIt`, if the status is `Detection` *and* the invoke-SD flag is set, the sensitive detector is invoked (lines 559–560):

$$f_{\text{status}} = \text{Detection} \ \wedge \ f_{\text{InvokeSD}} \implies \text{InvokeSD(step)}. \tag{2}$$

The flag is enabled in the physics list via `G4OpticalParameters::SetBoundaryInvokeSD(true)`. Consequently the sensitive-detector `ProcessHits` fires *only* for genuine photoelectrons (it tests that the step energy deposit is positive), counts one photoelectron per call, and kills the photon (the SiPM is opaque). This is a clean separation of concerns: the physics of the photocathode lives in the optical surface, and the sensitive detector only books hits.

### 2.12 Per-event optical accounting

Two per-event counts recur throughout this report. Write $N_{\text{pe}}^{(e,s)}$ for the number of photoelectrons detected in SiPM $s$ during event $e$, and $N_{\gamma}^{(e)}$ for the number of optical photons generated in the slab during event $e$ (scintillation plus Čerenkov), irrespective of their later fate.

A thread-local diagnostic record (`OpticalDiag`) accumulates, per event: scintillation photons, Čerenkov photons, terminated photons, detected photoelectrons, and photons that escaped to the world. The primary muon's first-entry position in the slab is captured on the first step inside the scintillator logical volume and stored in slab-local coordinates. At end of event the sensitive detector writes one row per (event, SiPM), *including zero-detection rows*, so the sample is unbiased in the muon interaction position. Each row carries the entry position, the event's produced-photon count (the efficiency denominator), the detected count in that SiPM, the first arrival time, the mean and RMS detected wavelength, and the track weight. This one tree supports per-SiPM, combined, and spatially-binned collection-efficiency analyses.

Figure 4 shows a head-less event display of a single vertical muon in the glossy-wrapped plate (config 1). For legibility the display uses a 30 MeV muon—just below the Čerenkov threshold of the scintillator—with the scintillation yield rescaled, so that the individual optical tracks can be followed: photons are generated along the muon path, propagate by total internal reflection and specular reflection off the wrapper, and terminate on the SiPM patches at the four edges.

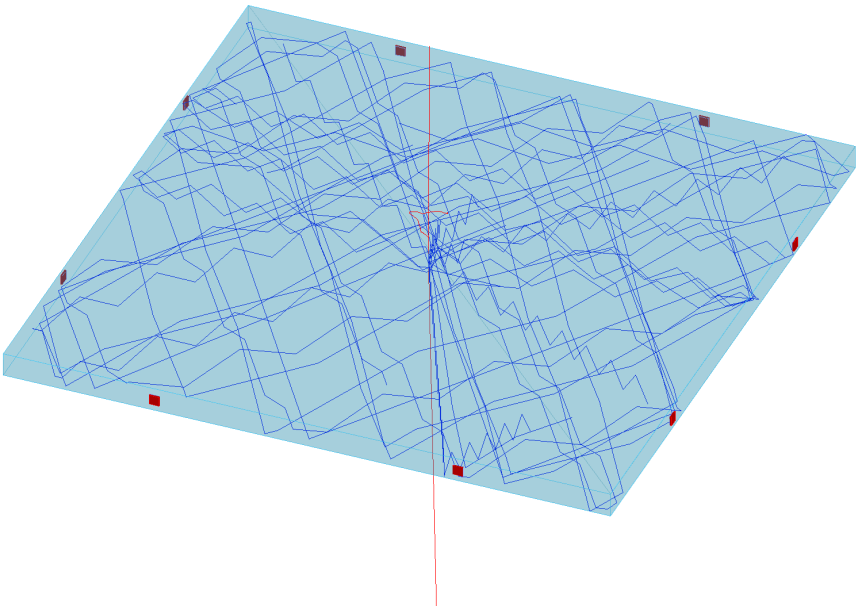


**Figure 4:** Head-less (`TSG_OFFSCREEN`) event display of one vertical muon in the glossy-wrapped plate with the config-1 layout, on a white background. The red line is the muon and the blue lines are optical photons (here purely scintillation, since the muon is below the Čerenkov threshold); the plate and its reflective wrapper are drawn as translucent surfaces with their edges outlined, and the SiPMs are opaque red. Both the muon energy (30 MeV, below the Čerenkov threshold) and the photon yield are chosen for visualisation only, so that individual tracks are resolved rather than merging into a saturated blob. The display is generated reproducibly by `macros/muon_vis.mac`.

## 3 Matrix simulation

### 3.1 Beam, irradiation patches, and efficiency definition

The primary source is a vertical muon pencil beam: a 50 mm×50 mm square beam of 4 GeV $\mu^-$ normally incident on the plate, with the beam centre placed at three positions, each irradiating a 50 mm×50 mm patch lying *entirely on* the polished plate. With the plate spanning $[-125, 125]$ mm in both $x$ and $y$, the patch centres are

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

so the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$ respectively (report frame: $+y$ is the top, $-x$ the left). This places the edge and corner regions on the bottom ($-y$) edge, opposite the single instrumented face of config 3 (top, $+y$), as in Fig. 3 of Ref. [1]. It also matches the 50 mm×50 mm active area of the muon coincidence detector (MCD) that defines the regions of interest in Ref. [1], and keeps the irradiated area—and hence the denominator of Eq. 3—equal at all three positions. Each (config, reflector, position) runs 1000 events after a single `/run/initialize`. The matrix therefore comprises

$$4\ \text{configs} \times 4\ \text{reflectors} \times 3\ \text{positions} = 48\ \text{files}, \qquad 48 \times 1000 = 48\,000\ \text{events}.$$

Runs used 8 threads on an 8-thread (4-physical-core) workstation. The optional per-slab optical track-length flux map is disabled for the matrix (it is retained as a separate diagnostic); the collection efficiency is reconstructed entirely offline from the `sipmHits` tree at any desired binning, so the map is not needed.

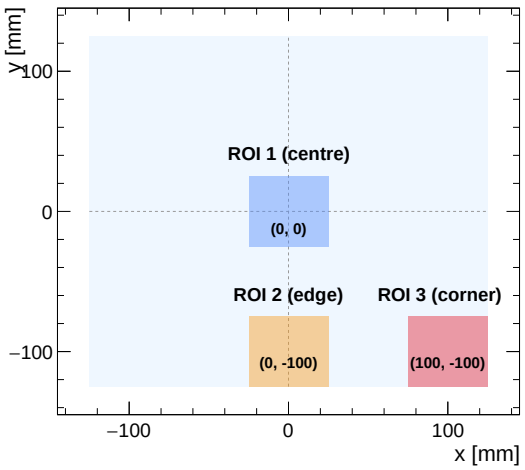



**Figure 5:** Plan view of the three irradiation sites (regions of interest) on the 250 mm×250 mm plate, in the report frame ($+y$ top, $-x$ left). Each coloured square is the 50 mm×50 mm footprint of the vertical muon beam: ROI 1 (centre, blue) at $(0, 0)$, ROI 2 (edge, orange) at $(0, -100)$, and ROI 3 (corner, red) at $(100, -100)$. The edge and corner sites lie on the bottom ($-y$) edge, opposite the single instrumented face of config 3.

The collection efficiency is defined, for a set $\mathcal{S}$ of SiPMs and an event $e$, as

$$\varepsilon = \frac{\sum_{e} \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_{e} N_{\gamma}^{(e)}}, \tag{3}$$

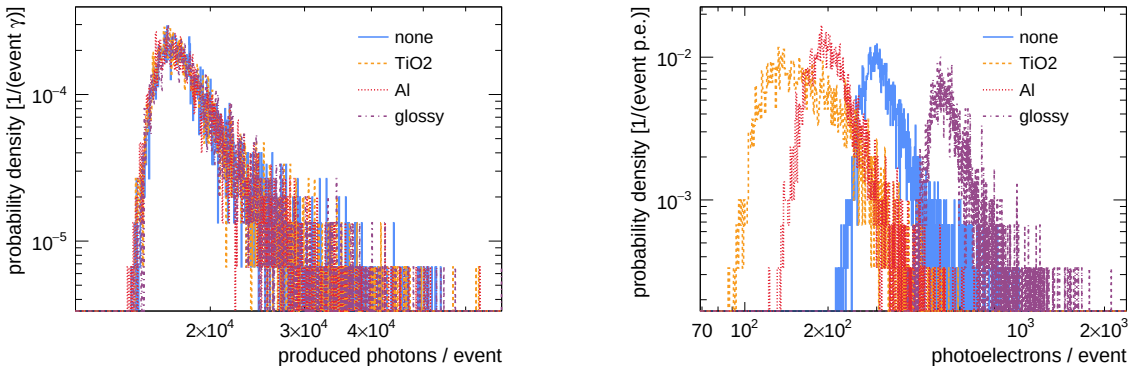
where the outer sum runs over all simulated events and the inner sum over the SiPM set $\mathcal{S}$. The all-SiPM sum over the three beam positions is the per-position figure of merit. A *plate-averaged* efficiency is the appropriate global figure of merit, but the three patches sample only part of the plate; it is therefore obtained from the uniform whole-plate irradiation of Sec. 4 and reported there.

For presentation, all position-resolved results (the maps of Fig. 5 and the whole-slab maps of Sec. 4) are shown in the report frame with the top edge at $+y$, the bottom at $-y$, the left at $-x$ and the right at $+x$. The internal simulation frame is this view rotated by 90° clockwise; the primary muon entry points and the SiPM markers are mapped into the report frame by $(x, y) \to (y, -x)$ before plotting. This rotation is a rigid change of view and leaves every scalar result (yields, spectra, efficiencies) unchanged; it only fixes the orientation of config 3 (top edge) and of the edge/corner irradiation sites (bottom edge).

### 3.2 Photon yield and photoelectron spectra

Figure 6a shows the per-event optical-photon yield (the denominator of Eq. 3) for the four reflectors. The mean yield is essentially independent of the reflector—about 19 000 photons per event—which is the correct physical behaviour: the reflector changes how photons are transported, not how many are produced. The small spread reflects event-to-event variation in the muon path length.

Figure 6b shows the corresponding photoelectron spectra. Here the reflector has a clear effect: the config-1 centre mean runs from 142 $N_{\mathrm{pe}}$/event for $TiO_2$ and 227 $N_{\mathrm{pe}}$/event for aluminium up to 347 $N_{\mathrm{pe}}$/event for the air-gap bare plate and 599 $N_{\mathrm{pe}}$/event for glossy paper. In particular the bare curve is *not* a near-zero peak but a broad distribution comparable to the reflective wrappers—the direct signature of its near-lossless TIR transport.




**(a)** Yield (produced photons/event).

**(b)** Photoelectrons/event.

**Figure 6:** Per-event distributions for config 1, plotted as a probability density (unit area, 1 event per unit of the abscissa) with logarithmic $x$- and $y$-axes. The production yield (left) is reflector-independent, as it must be; the detected photoelectron count (right) depends strongly on the wrapper. The log $y$-axis is needed because the four curves have the same event count but very different widths and heights: the narrow, low-occupancy $TiO_2$ distribution and the broad, high-occupancy bare and glossy distributions cannot both be resolved on a linear scale.

The efficiency scale is set by the SiPM PDE and the coupling: of order 19 000 photons are produced and the best wrapper delivers $\approx 599$ photoelectrons, i.e. $\approx 3\%$. Given a PDE of $\sim 0.4$ and the geometric acceptance of a side-mounted sensor, this is consistent.

### 3.3 Wavelength of detected photons

Figure 7 shows the wavelength distribution of detected photoelectrons, weighted by their count, for the four reflectors. The distribution peaks near 460 nm for the reflective wrappers and the bare plate (measured config-1 centre means of 455.2 nm for aluminium and 459.9 nm for the bare plate), i.e. slightly red-shifted with respect to the 425 nm emission peak. This is the expected consequence of self-absorption: the shorter-wavelength part of the emission overlaps the absorption band and is attenuated over the multi-bounce path, so the surviving and detected spectrum is red-shifted. With the air-gap model the bare plate now has as many detected photons as the reflective wrappers, so its spectrum is equally well defined.

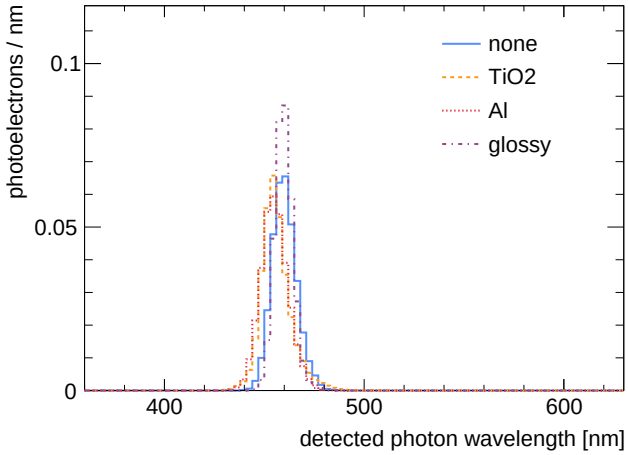



**Figure 7:** Wavelength distribution of detected photoelectrons (config 1), weighted by detected count. The reflective wrappers give a distribution peaking near 460 nm, red-shifted from the 425 nm emission peak by self-absorption over the multi-bounce path.

Figure 8 compares the detected-photon spectrum of the four configurations for each wrapper. Configurations 1–3, in which the SiPMs view the tile directly, all detect the same blue scintillation spectrum: the *mean* wavelength is essentially independent of the number of sensors, moving only from 459.6 nm (glossy, config 1) to 460.3 nm (config 3), the small increase reflecting the slightly longer collection path of the single-sided layouts. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, so the detected spectrum is shifted to 485.5 nm (glossy) — a green-shift of some 25 nm. The same shift appears for every wrapper, including the bare tile (which sits somewhat lower, near 474 nm), confirming that it originates in the fibre wavelength shifter rather than in the wrapper.

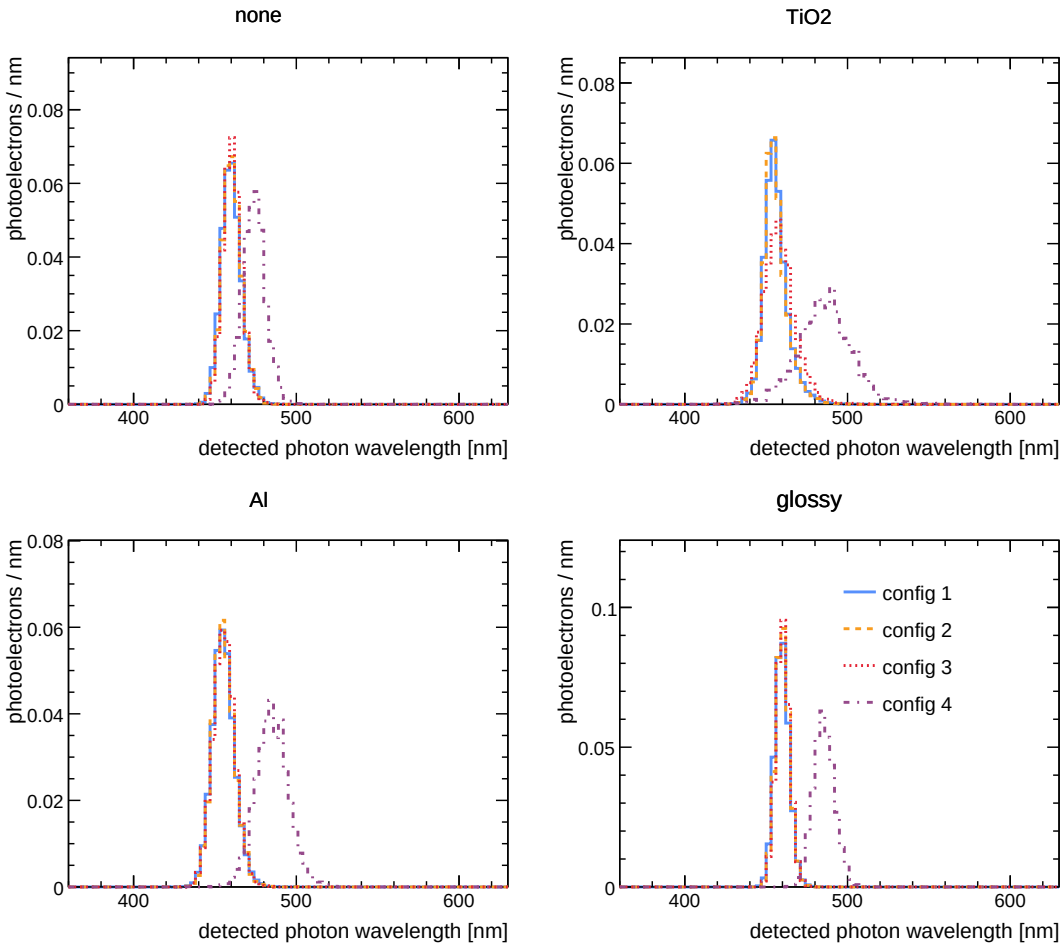



**Figure 8:** Detected-photon wavelength distribution, weighted by photoelectron count, for each wrapper (panel) and each configuration (curves), using the pooled data of the three irradiation positions. Configurations 1–3 follow the 460 nm blue scintillation spectrum regardless of sensor count, whereas the WLS-fibre readout of config 4 is green-shifted to $\approx 485$ nm by the Y-11(200) emission band.

### 3.4 Spatial and directional dependence

The matrix scan separates the three 50 mm×50 mm patches (centre, edge, corner) precisely so that the response can be resolved by position. The tempting single-coordinate summary is the distance $r$ of the patch from the plate centre, but $r$ is blind to azimuth, and the shortcoming is decisive here: configs 3 and 4 instrument a *single* face, so the collection depends on the *direction* from which the plate is irradiated, not merely on the distance. All three patches also lie on the half of the plate opposite the instrumented face of the single-sided layouts, so their distances are not even monotonic in the proximity to a sensor. We therefore keep the three patches separate and compare them directly.

Figure 9 groups the three patches by layout (one panel per config, four reflector curves) and Fig. 10 groups them by reflector (one panel per wrapper, four layout curves). Because every patch lies fully on the plate and all three have equal area, the produced-photon denominators are equal and the three points within each curve are directly comparable. The curves are plotted on a logarithmic $y$-axis, which resolves the weaker $TiO_2$ wrapper alongside the others.

The conspicuous feature is that the bare plate and the reflective wrappers are nearly *flat* across the three patches: for glossy paper and aluminium the efficiency varies by at most 6 % in every layout, and for the air-gap bare plate by at most 18 % (and by 5 % or less in the four-sided layouts 1 and 2). Only the diffuse $TiO_2$ paint shows a clear position dependence, varying by a factor $\approx$1.7–2.0. This is the expected behaviour of a closed cavity: after the first few bounces the

light no longer remembers where the muon crossed the plate, so the sensor coverage, and not the irradiation point, sets the collection. The specular foil and glossy paper homogenise the response directly; the air-gap bare plate homogenises it more slowly, but because its bulk attenuation is weak the TIR random walk mixes the light just as thoroughly by the time it reaches the edges.

These three patches lie near the *instrumented* half of the plate, so the comparison above captures the position dependence within the well-lit region. The response over the *whole* plate is resolved in Sec. 4.3; there the bare and specular wrappers are confirmed to be almost uniform, while $TiO_2$ retains the strongest (and now centre-depressed) position dependence.

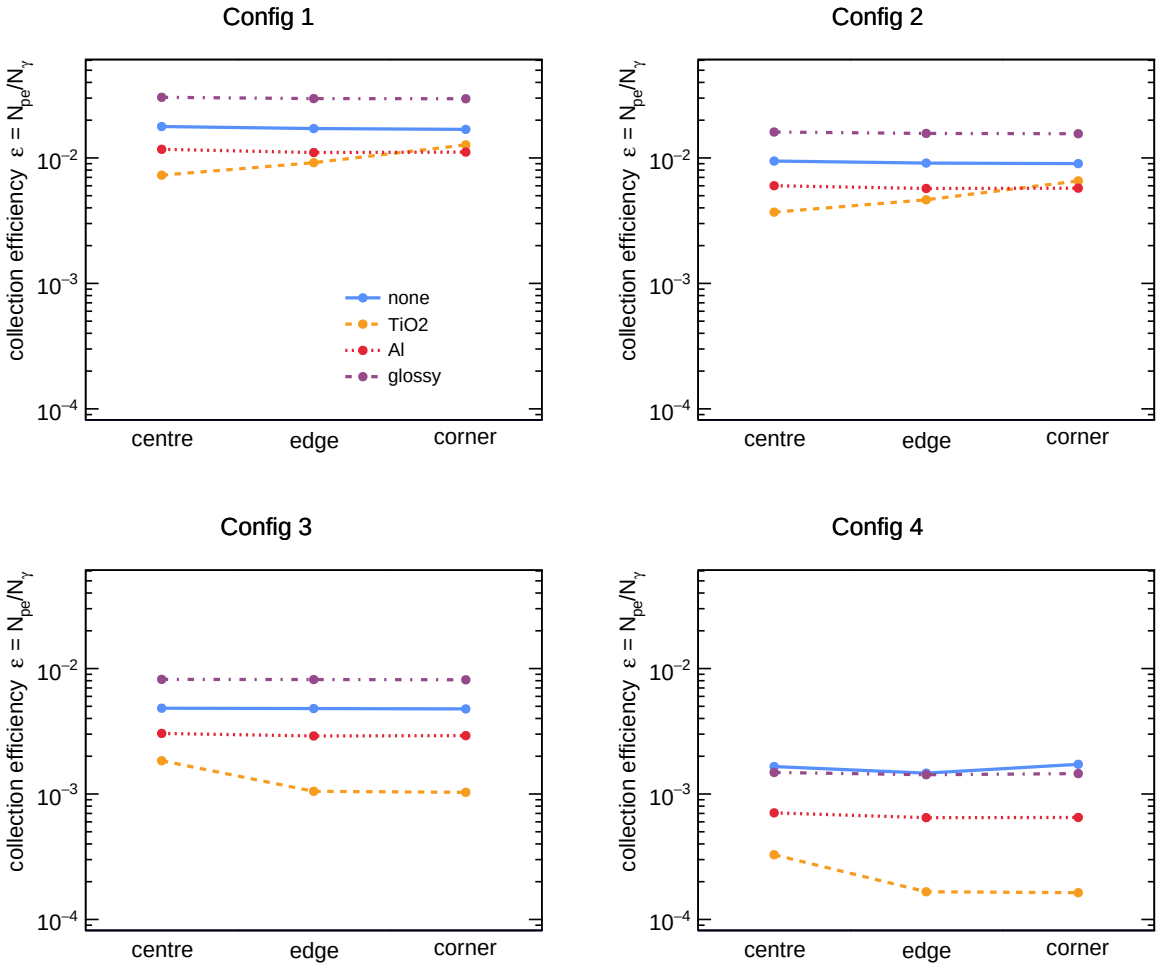



**Figure 9:** Collection efficiency at the three irradiation patches (centre, edge, corner), one panel per detector layout, with the four reflector curves (logarithmic $y$-axis). All three patches have equal area and lie fully on the plate, so the points are directly comparable. The bare plate and the specular wrappers are nearly flat across the three patches in every layout; only the diffuse $TiO_2$ paint shows a clear position dependence.

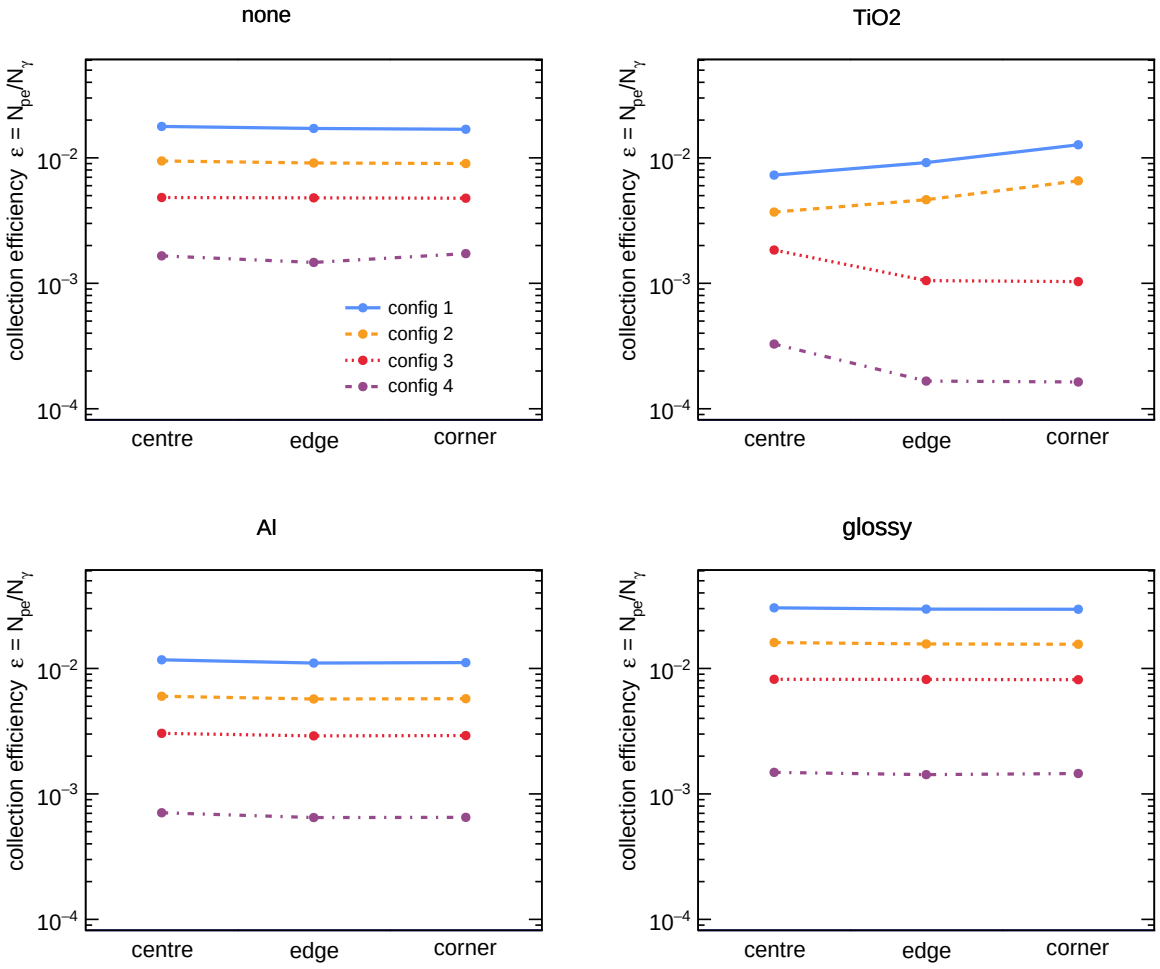



**Figure 10:** The same data grouped by reflector (one panel per wrapper), with the four layouts as curves. Configs 1 and 2 (sensors on every face) are nearly flat for every wrapper; in the single-sided layouts (3 and 4) the diffuse $TiO_2$ drops from centre toward the edge and corner patches, which sit on the far side of the plate from its sensors, while the bare and specular wrappers remain flat.

Figure 11 resolves the directional response most directly. For config 1 (the layout with the most sensors) it shows the *share* of the collected light seen by each of the eight sensors, in one panel per reflector, with the three irradiation patches as grouped bars (dashed line: the uniform expectation of 1/8). The difference between the wrappers is instructive. For the air-gap bare plate, aluminium and glossy paper the light is well mixed: at the corner patch the largest and smallest single-sensor shares differ by less than a factor of about 1.5, and the specular wrappers are flatter still. Only the diffuse $TiO_2$ paint concentrates the light strongly: at the corner patch the two sensors on the faces bordering the illuminated corner together collect 75 % of the photoelectrons (about three times the uniform share each), while the remaining six share the rest. The total efficiency is nearly the same for all three patches (Table 3), so the cavity keeps the *amount* of collected light constant while rearranging its spatial division. A radius-only summary averages these distinct per-sensor patterns into a single number and hides precisely this information.

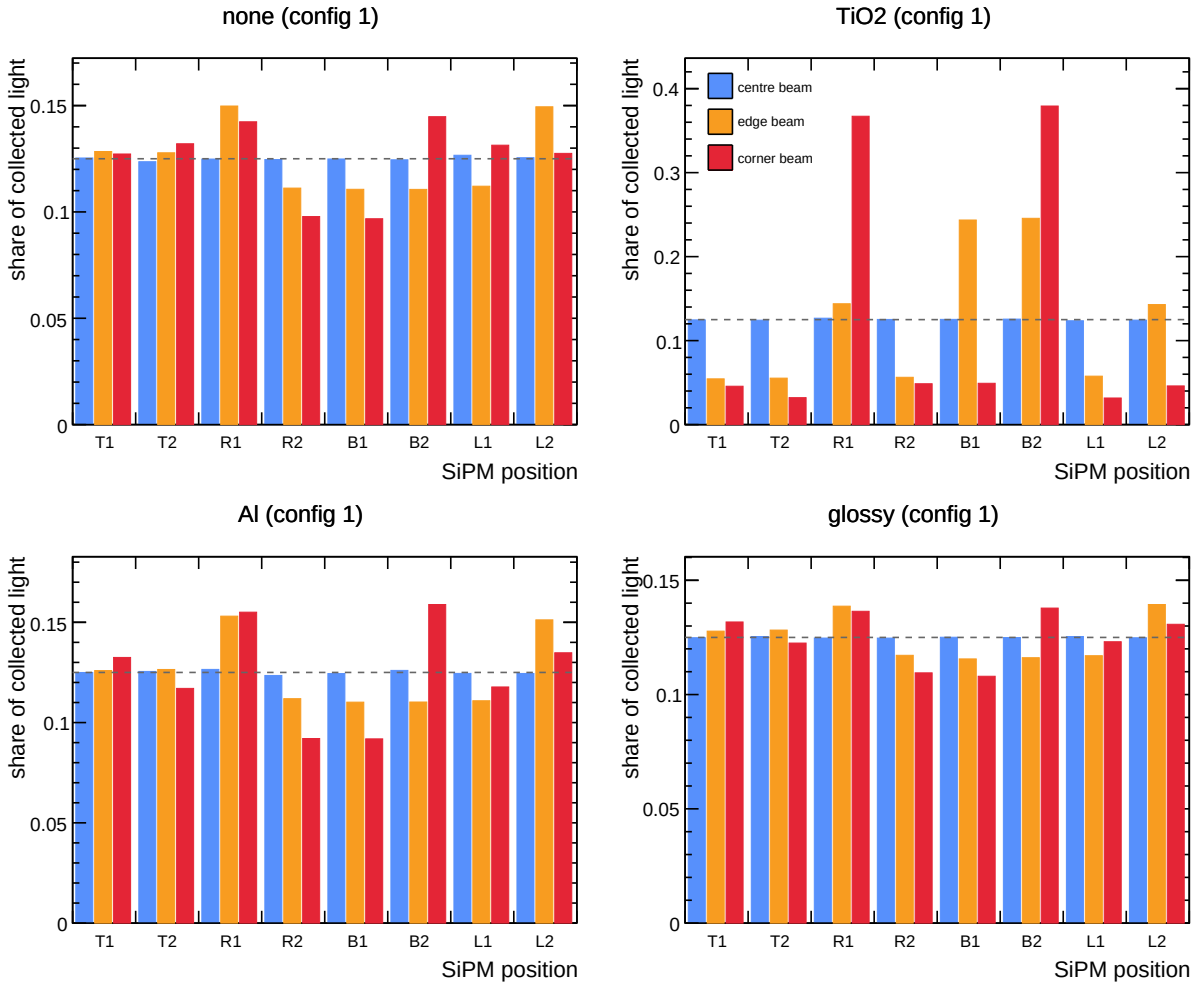



**Figure 11:** Direction-aware light sharing, config 1, one panel per reflector. Each group of bars is one sensor (x axis) and the three colours are the irradiation patches (blue: centre, orange: edge, red: corner); the height is that sensor's share of the light collected at that patch, and the dashed line marks the uniform expectation 1/8. The bare plate and the specular wrappers spread the light almost uniformly over all eight sensors, whereas the diffuse $TiO_2$ paint concentrates it on the two sensors nearest the beam.

Table 3 lists the absolute centre/edge/corner efficiencies for config 1; the fully position- and direction-resolved maps over the *entire* plate are given in Sec. 4.

**Table 3:** Position dependence of the collection efficiency, config 1 (all SiPMs). Each beam patch is 50 mm×50 mm and lies fully on the plate, so the produced-photon denominators are equal.

| Reflector | Centre | Edge | Corner |
|---|---|---|---|
| Bare | $1.78 \times 10^{-2}$ | $1.71 \times 10^{-2}$ | $1.69 \times 10^{-2}$ |
| $TiO_2$ | $7.3 \times 10^{-3}$ | $9.2 \times 10^{-3}$ | $1.27 \times 10^{-2}$ |
| Al | $1.17 \times 10^{-2}$ | $1.10 \times 10^{-2}$ | $1.11 \times 10^{-2}$ |
| Glossy | $3.04 \times 10^{-2}$ | $2.97 \times 10^{-2}$ | $2.96 \times 10^{-2}$ |

### 3.5 Photoelectron spectra across configurations and positions

Figure 12 collects the per-event photoelectron spectra over the whole matrix scan in the layout of Fig. 10 of Rao et al. [1]: the three irradiation regions (centre, edge, corner) are the rows and the detector configurations are the columns, with the four reflector treatments overlaid in every cell. Two adaptations are made. First, the voltage pulse-height spectra of the measurement are replaced by the simulation's number of photoelectrons per event, the quantity that is directly

comparable between our model and the experiment. Second, the fourth configuration (single SiPM + WLS fibre clamp) is added as a fourth column, so that all four layouts appear in one figure on a common logarithmic axis (panel (b)) rather than in a separate figure.

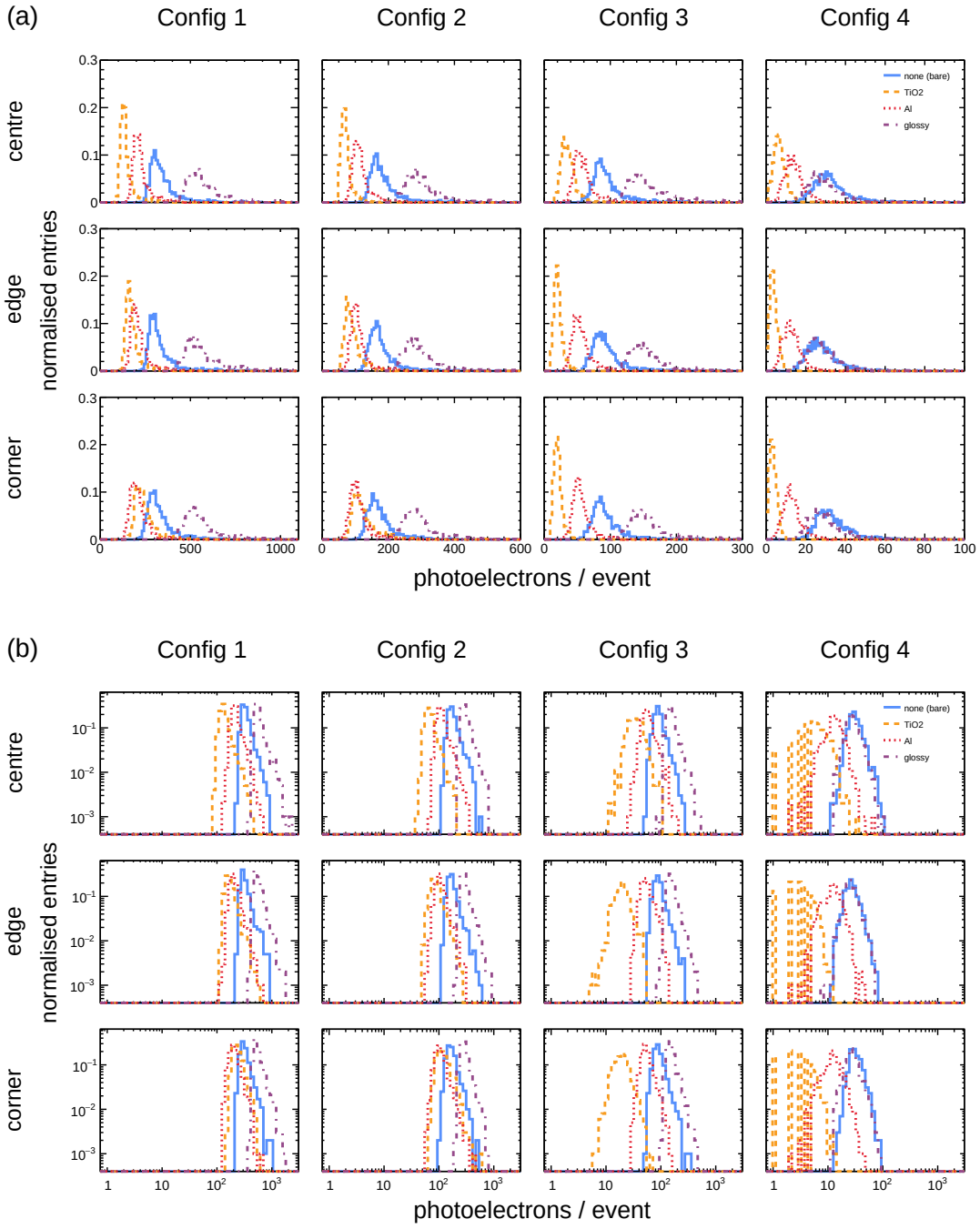



**Figure 12:** Photoelectron spectra in the layout of Fig. 10 of Rao et al. [1]: rows are the irradiation regions (centre, edge, corner) and columns the detector configurations (1–4, with the WLS-clamp layout as the fourth column). Each cell overlays the four reflector treatments. The histograms are normalised entry distributions (each cell is scaled so its entries sum to unity). The *upper* grid (a) uses linear axes and the *lower* grid (b) logarithmic axes; both share the same reflector colours and styles and the same units. Panel (a) gives each column its own linear abscissa, truncated at that configuration's 99.5th percentile (rounded up) and binned linearly, so that the narrow high-density core of every layout is resolved. A single shared linear abscissa would instead collapse the WLS-clamp layout (config 4), which collects roughly an order of magnitude fewer photoelectrons than the direct-coupled layouts, into the first few percent of the axis. Panel (b) shows every cell over its full distribution on a single shared logarithmic abscissa with logarithmic bins, so the four layouts remain directly comparable there. Zero-photoelectron events form a pulse below threshold and lie off the abscissa. The wrappers shift and broaden the spectrum from the low, narrow $TiO_2$ distribution to the wide, bright glossy one, with the air-gap bare plate close to glossy and the foil in between; apart from $TiO_2$ the spectra are nearly independent of position.

The figure condenses the trends established above. At a fixed position the reflector ordering $TiO_2 < Al <$ bare $<$ glossy holds in every direct-coupled cell, with $TiO_2$ the lowest and glossy the highest; the air-gap bare plate sits between the foil and glossy paper, its spectrum as broad and well populated as the reflective wrappers. The position dependence is wrapper-specific: for the diffuse $TiO_2$ paint the corner spectrum is shifted highest in the four-sided layouts (a factor $\approx 1.7$ in the mean, config 1) but the centre is highest in the single-sided layouts 3 and 4, whereas the bare and specular wrappers are essentially degenerate across positions, reflecting the flat response found above. This is the same reflector-versus-position interplay summarised in Table 3 and the efficiency maps, presented here as the pulse-height distributions that a threshold-based read-out would actually see.

## 4 Whole-slab irradiation

### 4.1 Plate-averaged collection efficiency

The matrix scan pools only three 50 mm×50 mm patches, all on the $+x$ half of the plate, so the efficiency it yields depends on the choice of patches and is *not* a plate average. The whole-slab run removes that bias: the GPS plane is expanded to $250 \times 250$ mm (`halfx` = `halfy` = 125 mm) and 25 000 events are generated per (config, reflector), so the primary muon's first-entry position is sampled *uniformly* over the entire plate and the pooled ratio $\sum N_{\mathrm{pe}} / \sum N_{\gamma}$ is the *plate-averaged* collection efficiency—the appropriate global figure of merit. Figure 13 shows it for every (config, reflector) pair, and Table 4 lists the numbers.

The measured ordering of Ref. [1] is only partly reproduced. Glossy paper is the brightest wrapper in every layout and $TiO_2$ paint the weakest, as the experiment finds, but the bare plate is *not* the weakest collector: with its true air gap it is the *second*-brightest configuration in configs 1–3 and the brightest in config 4. The resulting orderings are

$$TiO_2 < Al < \text{bare} < \text{glossy} \quad (\text{configs } 1\text{–}3), \qquad TiO_2 < Al < \text{glossy} < \text{bare} \quad (\text{config } 4),$$

so the spread runs over two orders of magnitude from $TiO_2$ to glossy, and the bare plate sits near the top rather than at the bottom. The reason is that a polished plate in air is a nearly lossless TIR light guide (Sec. 2.7): without a rough or optically-contacting surface the trapped light is efficiently transported to the edge sensors, so the bare collection is comparable to—and, for the grease-coupled foil whose reflectivity is only $\approx 90\%$, higher than—the aluminium wrapper. The efficiency scales roughly with the photosensitive area: the eight-SiPM layout (config 1) is highest, four SiPMs on four faces (config 2) roughly halve it, and concentrating the two sensors of config 3 on a single face costs a further factor. The single-SiPM WLS-fibre layout (config 4) collects substantially less than the direct-coupled layouts: only the fraction of light that reaches the fibre face is wavelength-shifted and guided to the sensor, so its efficiency sits roughly a factor of three to six below config 3. It is also the only layout in which the bare plate is the single most efficient configuration.

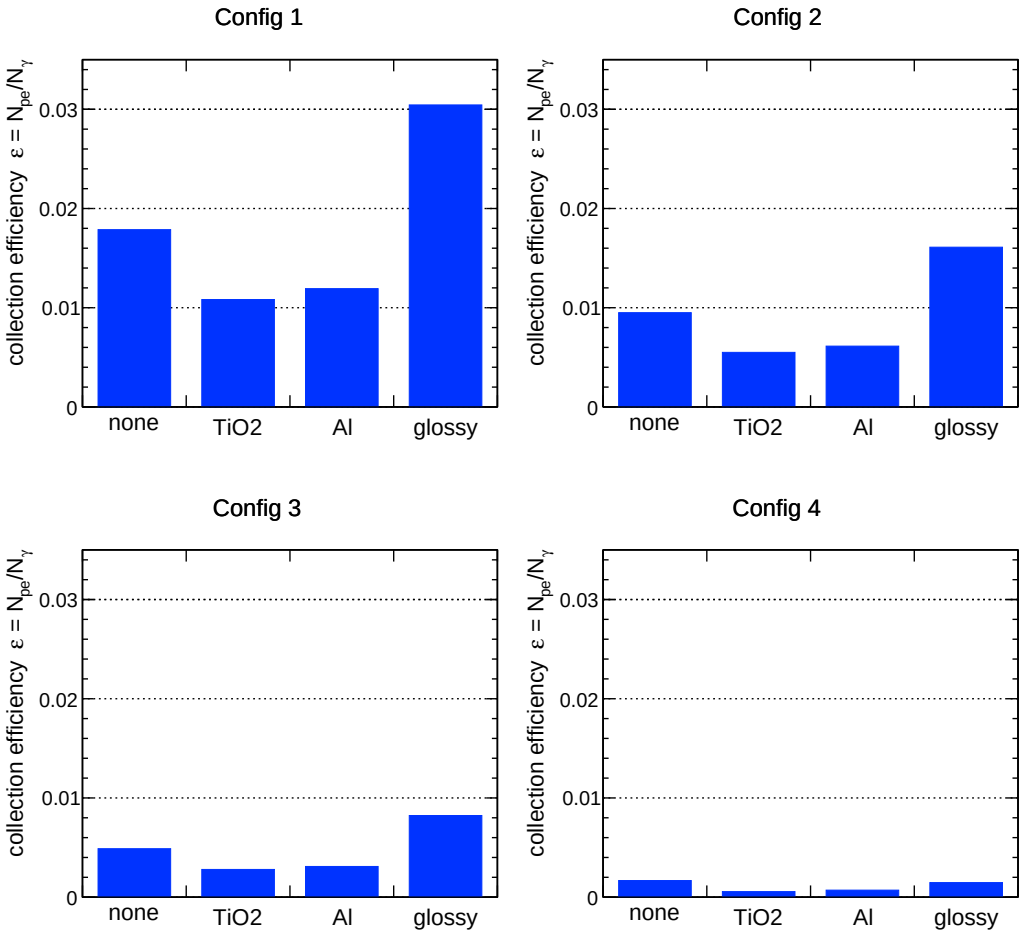



**Figure 13:** Plate-averaged collection efficiency $\varepsilon = \sum N_{\mathrm{pe}} / \sum N_{\gamma}$ for the four reflector treatments and four detector layouts, from the uniform whole-plate irradiation (25 000 events per panel). Glossy paper is brightest and $TiO_2$ weakest in every layout, but the air-gap bare plate is second only to glossy (and first in the WLS layout), not the weakest.

**Table 4:** Plate-averaged collection efficiency (all SiPMs, uniform whole-plate irradiation; the pooled ratio of Eq. 3 over all events). The statistical uncertainties are below the quoted digits.

| Layout | Bare | $TiO_2$ | Al | Glossy |
|---|---|---|---|---|
| Config 1 (8 SiPMs) | $1.79 \times 10^{-2}$ | $1.08 \times 10^{-2}$ | $1.19 \times 10^{-2}$ | $3.04 \times 10^{-2}$ |
| Config 2 (4 SiPMs) | $9.50 \times 10^{-3}$ | $5.50 \times 10^{-3}$ | $6.13 \times 10^{-3}$ | $1.61 \times 10^{-2}$ |
| Config 3 (2 SiPMs) | $4.89 \times 10^{-3}$ | $2.79 \times 10^{-3}$ | $3.09 \times 10^{-3}$ | $8.23 \times 10^{-3}$ |
| Config 4 (1 SiPM, WLS) | $1.68 \times 10^{-3}$ | $5.60 \times 10^{-4}$ | $7.09 \times 10^{-4}$ | $1.47 \times 10^{-3}$ |

### 4.2 Full-plate maps

The same whole-slab events that give the plate average above also resolve the response by entry position. Reconstructing the efficiency from the `sipmHits` tree in $5 \times 5$ mm cells gives the field

$$\varepsilon(x, y) = \frac{\sum_{e\,:\,(x_e, y_e) \in \mathrm{cell}} \; \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_{e\,:\,(x_e, y_e) \in \mathrm{cell}} \; N_{\gamma}^{(e)}},$$

where the sums run over the whole-slab events whose primary-muon first-entry position $(x_e, y_e)$ falls inside the cell, i.e. identically to Eq. 3 but binned in the entry position. The whole-slab maps therefore fill in the gaps left by the three-patch scan.

Figure 14 shows the complete set of full-plate maps for config 1, one panel per reflector. For the bare plate the plate-averaged efficiency is $\varepsilon = 1.79 \times 10^{-2}$, consistent with the three-patch bare values of Table 3, and the map is nearly *flat*: the mean within $r < 60\,\mathrm{mm}$ ($1.77 \times 10^{-2}$) differs from that in the outer region ($1.79 \times 10^{-2}$) by under 2 %. This is the direct consequence of the TIR-guide picture: because bulk attenuation is weak, photons born at the centre random-walk to a sensor-bearing edge with little loss, so the entry position barely matters.

The other panels show the same weak position dependence, with $TiO_2$ as the one exception. Aluminium is essentially flat ($1.17 \times 10^{-2}$ for $r < 60\,\mathrm{mm}$ versus $1.20 \times 10^{-2}$ beyond) and glossy paper flatter still (3.04 versus $3.05 \times 10^{-2}$). $TiO_2$ is the only wrapper with a clear gradient, and it runs the other way: the centre ($7.7 \times 10^{-3}$) is *dimmer* than the edge and corner ($1.15 \times 10^{-2}$), a factor of $\sim$1.5. Diffuse recycling couples light into the sensors most efficiently when the source sits near a painted edge, and least when it is in the middle of the plate, whereas the specular and TIR transports erase the entry position altogether.

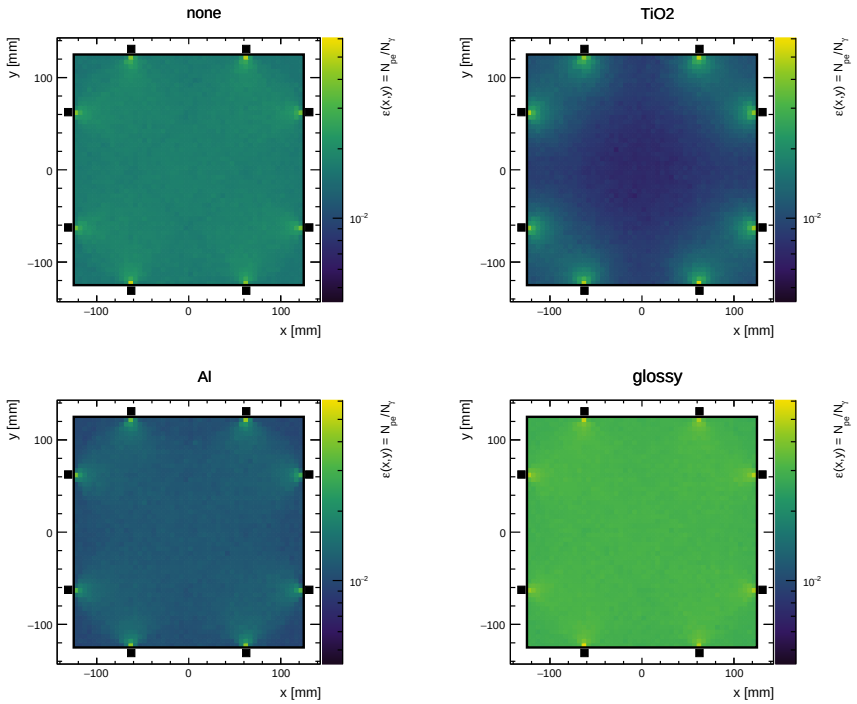



**Figure 14:** Full-plate collection-efficiency maps for config 1, from 25 000 whole-slab events per panel ($5 \times 5$ mm cells, on a logarithmic colour scale shared by the panels). The bare plate (top left) and the specular wrappers are almost flat; $TiO_2$ (top right) is the weakest and the only one with a clear gradient, dimmest at the centre and brightest toward the edges and corners. The bold rectangle is the 250 mm×250 mm slab outline and the black squares mark the eight SiPM positions.

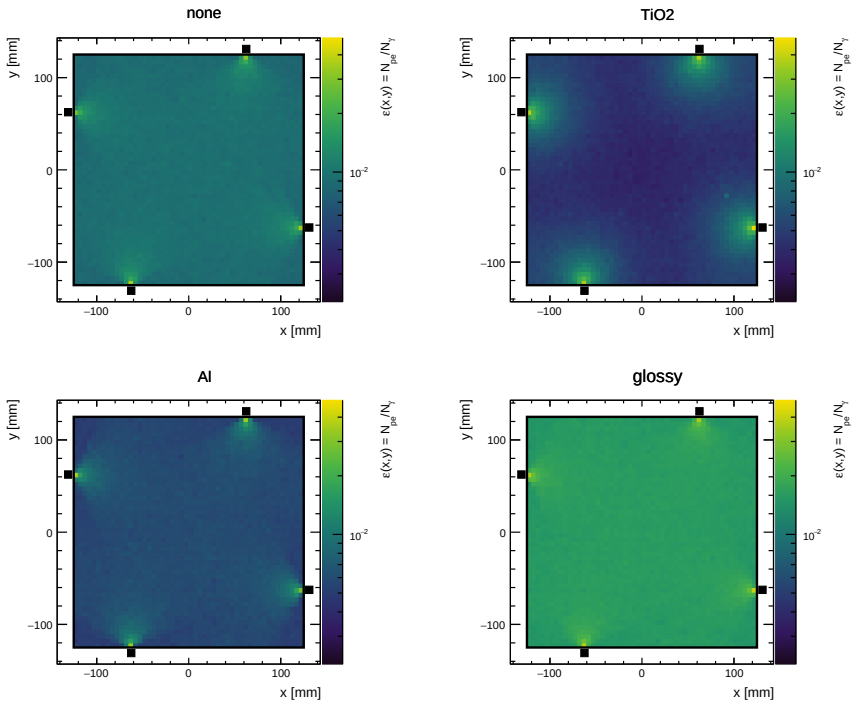



**Figure 15:** Full-plate collection-efficiency maps for config 2 (four SiPMs, one per lateral face at the config-1 offsets), otherwise identical to Fig. 14. The reflector ordering and the near-flat response are the same, with each efficiency about half the config-1 value.

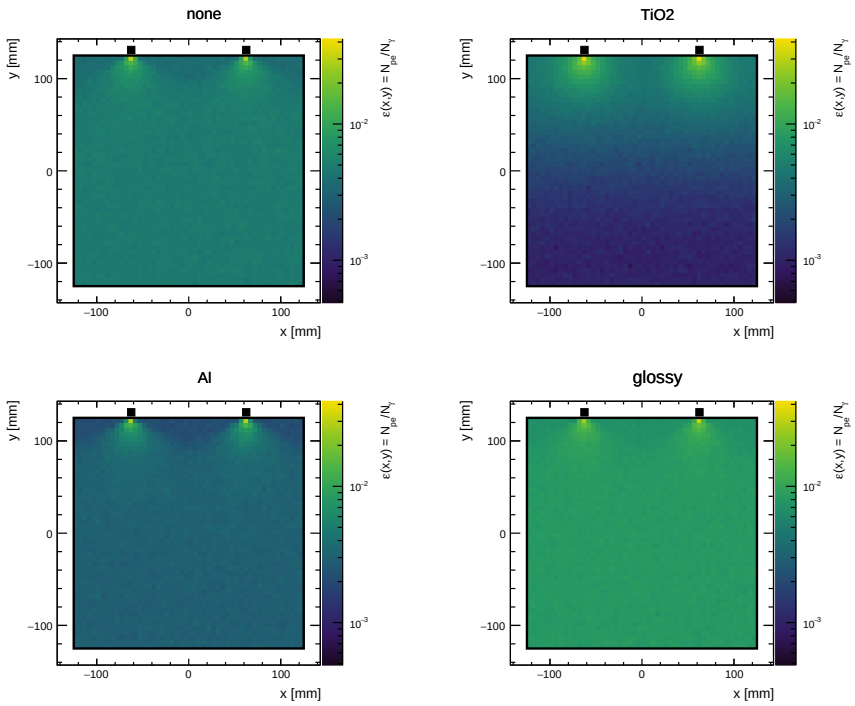



**Figure 16:** Full-plate collection-efficiency maps for config 3 (two SiPMs, both on the top $+y$ face). The efficiencies are the lowest of the direct-coupled layouts; the $TiO_2$ panel shows the clearest top–bottom asymmetry, while the bare and specular wrappers stay nearly flat.

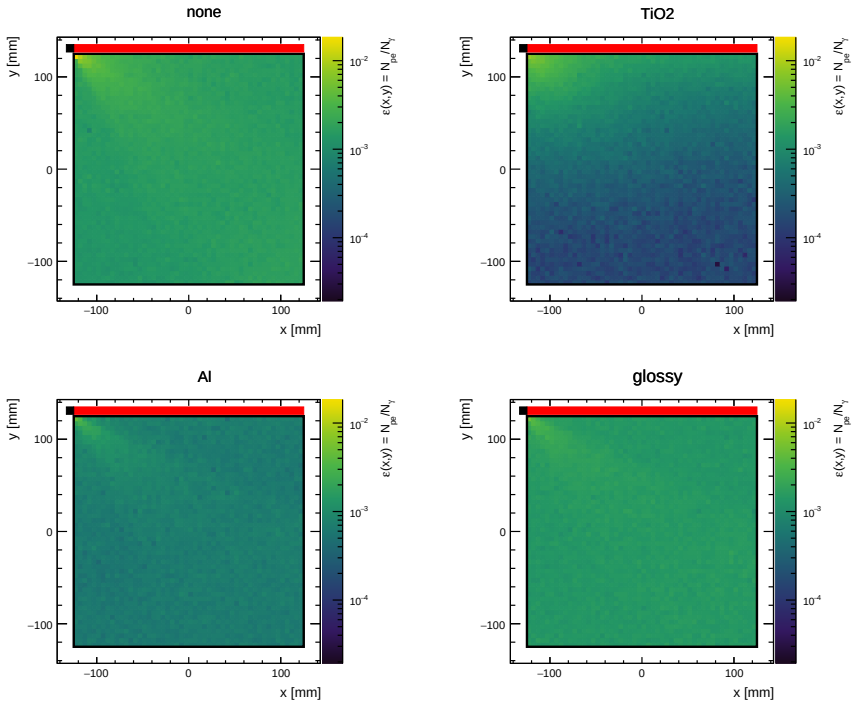



**Figure 17:** Full-plate collection-efficiency maps for config 4 (single SiPM + WLS fibre clamp). The lowest efficiencies of all four layouts appear here; the $TiO_2$ panel retains a centre-to-edge gradient, while the bare and specular wrappers are almost flat. The red bar along the top ($+y$) edge marks the WLS fibre clamp (the fibres run the full plate length along it) and the black square at its left end is the single read-out SiPM, so the fibres can be located relative to the read-out. Same report frame and binning as Figs. 14–16, each panel on its own logarithmic colour scale.

Figure 15 repeats the whole-slab scan for config 2 (four sensors, one on each lateral face at the config-1 offsets). Every configuration is weaker than its config-1 counterpart—roughly halved, as expected from the reduced sensor area—while the reflector ordering and the near-flat response are unchanged. The bare plate gives $\varepsilon = 9.5 \times 10^{-3}$, again flat ($9.4 \times 10^{-3}$ at the centre versus $9.5 \times 10^{-3}$ outside $r = 60\,\mathrm{mm}$); $TiO_2$ reaches $5.5 \times 10^{-3}$ with the same centre deficit (3.9 versus $5.9 \times 10^{-3}$), and Al and glossy reach $6.13 \times 10^{-3}$ and $1.61 \times 10^{-2}$, both nearly flat. The $TiO_2$ < Al < bare < glossy sequence and the near position-independence of the bare and specular wrappers are therefore robust against the sensor count: fewer sensors lower the overall light collection but do not change the physics of the wrapper.

Finally, Fig. 16 shows config 3, in which both sensors sit on the same (top, $+y$) face. The integrated efficiencies are the lowest of the direct-coupled configurations—bare $4.9 \times 10^{-3}$, $TiO_2$ $2.8 \times 10^{-3}$, Al $3.1 \times 10^{-3}$, glossy $8.2 \times 10^{-3}$—because concentrating both sensors on one face halves the solid angle available to photons born on the opposite side. The bare, Al and glossy panels remain nearly flat (to better than 4 % between the inner and outer regions), whereas the $TiO_2$ panel is brightest near the instrumented top ($+y$) edge and dimmest on the far ($-y$) side ($1.9 \times 10^{-3}$ at the centre versus $3.0 \times 10^{-3}$ outside $r = 60\,\mathrm{mm}$).

Figure 17 completes the set with the single-SiPM WLS-fibre layout (config 4). The absolute efficiencies are the lowest of all four layouts —bare $1.68 \times 10^{-3}$, $TiO_2$ $5.6 \times 10^{-4}$, Al $7.1 \times 10^{-4}$, glossy $1.5 \times 10^{-3}$—as expected from the small fibre acceptance, but the ordering is changed: here the bare plate is the most efficient configuration, just above glossy paper (the same reversal appears in the plate averages of Table 4). $TiO_2$ again shows the clear gradient ($3.5 \times 10^{-4}$ versus $6.1 \times 10^{-4}$, a factor 1.7), whereas the bare plate ($1.66 \times 10^{-3}$ vs. $1.68 \times 10^{-3}$) and Al and glossy (agreeing to better than 2 %) are almost flat, so even with a single readout face the TIR and specular transports erase the entry-point dependence.

Across all four layouts the two weakest wrappers are the same—$TiO_2$ then Al—and glossy paper is the brightest wrapper in configs 1–3, while in config 4 the bare plate overtakes glossy. The efficiency scales roughly with the collected photosensitive area, confirming that the readout layout sets the overall scale while the wrapper sets the ordering.

### 4.3 Whole-slab uniformity

The full-plate maps also expose an inhomogeneity that the three discrete patches of Sec. 3 can only sample. Because the whole-slab beam irradiates every point of the plate, this inhomogeneity can be quantified *continuously* and with far better statistics.

A purely radial contrast is deliberately *not* used. The read-out layouts have no continuous rotational symmetry—configs 1 and 2 are four-fold symmetric, configs 3 and 4 single-sided—and the plate is square, so the collection efficiency is not a function of radius alone. Beyond the inscribed circle ($r > 125\,\mathrm{mm}$) only the four corner wedges of the plate are populated, so a fixed-width radial bin would mix a shrinking range of directions and its radial profile $\varepsilon(r)$ would reflect the geometry of the square rather than the detector. The uniformity is therefore quantified directly in the plate plane $(x, y)$.

*Two complementary uniformity measures.* The whole-slab scan gives the collection efficiency as a *field* $\varepsilon(x, y)$, the position-binned image of Eq. 3. We characterise this field in two ways. (i) The *normalised map* $\varepsilon/\langle\varepsilon\rangle$, with $\langle\varepsilon\rangle$ the plate-averaged efficiency, is shown for each layout in Figs. 19–22 in 5 mm cells. It is a dimensionless field of unit mean that shows *where* the response is high or low and the spatial pattern of the asymmetry (for instance a top–bottom gradient), but *not* the absolute efficiency. (ii) The *two-dimensional contrast*

$$\kappa_{2\mathrm{D}} \;=\; \frac{\max_{\mathrm{cell}}\varepsilon}{\min_{\mathrm{cell}}\varepsilon}$$

collapses the field to a single number, the ratio of the brightest to the darkest region; it answers "how large is the spread" and is the quantity that ranks the layouts and reflectors. $\kappa_{2\mathrm{D}}$ is

evaluated on a coarser $5 \times 5$ grid of 50 mm cells, because on the 5 mm cells of the maps the per-cell statistics are too thin for a stable max/min—Poisson noise in the extreme cells would inflate it. The two measures therefore use different binning: the contrast quoted below is deliberately the *robust* dynamic range, and is smaller than the raw peak-to-trough range of the fine maps. No symmetry is assumed, so $\kappa_{2D}$ is meaningful for the single-sided layouts 3 and 4 as well as for the four-sided ones.

The direction-sensitive contrast $\kappa_{2D}$ ranks the reflectors identically in *every* layout,

$$\mathrm{TiO_2} \;>\; \mathrm{Al} \;>\; \text{bare} \;\gtrsim\; \text{glossy},$$

i.e. the diffuse $TiO_2$ paint is the *least* uniform wrapper and glossy paper the most uniform. This runs opposite to the naive expectation that the brightest wrappers should be the most inhomogeneous, and it follows directly from the flat responses found above: the specular and TIR transports mix the light so thoroughly that the entry position is largely erased ($\kappa_{2D}$ near unity), whereas the diffuse paint retains a pronounced positional response. Numerically the $TiO_2$ contrast grows from $\kappa_{2D} = 1.89$ and 2.52 in the four-sided layouts 1 and 2 to 8.3 and 16.8 in the single-sided layouts 3 and 4, whereas the bare plate stays at $\kappa_{2D} = 1.12$–$2.38$ and the specular wrappers at 1.06–1.48. For the bare plate and the two specular wrappers a muon crossing anywhere on the plate is therefore collected within about a factor of two of one at the most favourable point, whereas for $TiO_2$ in the single-sided layouts the corner-to-centre penalty exceeds an order of magnitude. Figure 18 collects all sixteen values as bar charts, one panel per layout; Table 5 lists the same values numerically, resolving the near-unity entries that the linear bar scale cannot.

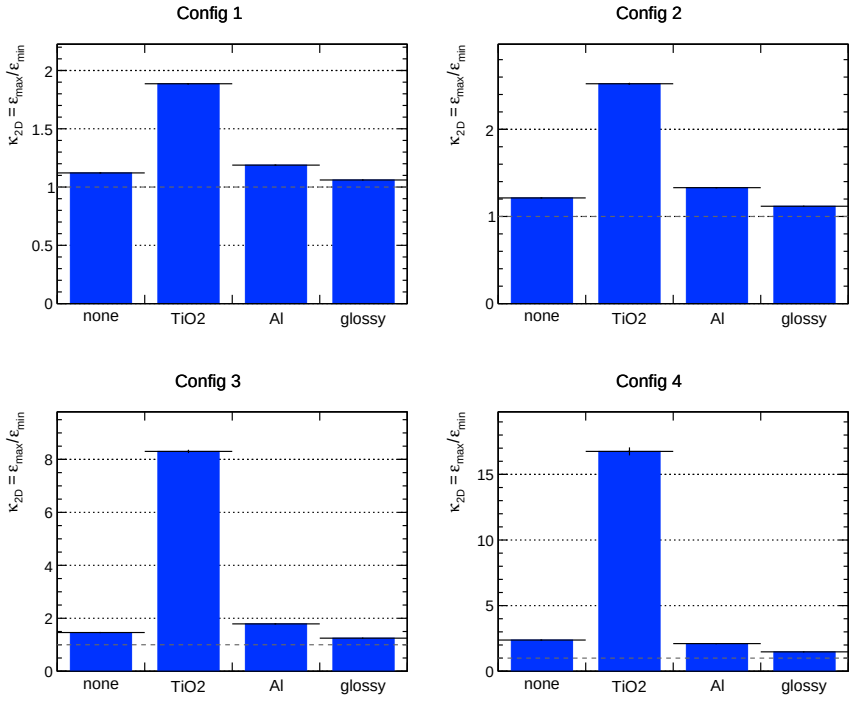



**Figure 18:** Full-plate position contrast $\kappa_{2D} = \varepsilon_{max}/\varepsilon_{min}$ over the 50 mm cells of the whole-slab map, one panel per layout and one bar per reflector (each panel on its own linear scale). This is the same kind of comparison as the three near-edge patches, but evaluated over the entire plate; the dashed line marks unity (a perfectly uniform response). $TiO_2$ shows by far the largest contrast in every layout, the Al and bare wrappers sit in between, and glossy paper stays closest to unity.

**Table 5:** Full-plate position contrast $\kappa_{2D}$ over the 50 mm cells of the whole-slab map, per layout and reflector, with the propagated Poisson uncertainty. These are the values plotted in Fig. 18; the table resolves the near-unity entries that the linear bar scale cannot, and shows the monotonic $TiO_2 > Al > \text{bare} \gtrsim \text{glossy}$ ordering in every layout. The number in parentheses is the uncertainty in the last digits of the preceding value.

| Reflector | Config 1 | Config 2 | Config 3 | Config 4 |
|---|---|---|---|---|
| Bare | 1.121(3) | 1.213(4) | 1.462(7) | 2.377(18) |
| $TiO_2$ | 1.886(6) | 2.523(11) | 8.302(62) | 16.75(31) |
| Al | 1.188(4) | 1.331(5) | 1.788(10) | 2.105(24) |
| Glossy | 1.061(2) | 1.117(3) | 1.249(4) | 1.477(12) |

The single-sided layouts are the most demanding. For config 4 the $TiO_2$ two-dimensional contrast is the largest of the entire study ($\kappa_{2D} = 16.8$): the front–back asymmetry of the diffusely wrapped plate is far stronger than any radial average reveals, although the bare and specular wrappers in the same layout stay below a factor of 2.4. A single-sided layout is simply not described by a radius, so only the full $(x, y)$ map—not a radial profile—carries the information.

Figures 19–22 show the same response in normalised form, $\varepsilon/\langle\varepsilon\rangle$, for each of the four layouts, on a shared *logarithmic* colour scale. A single linear scale cannot hold the range: for the diffuse $TiO_2$ paint the response spans well over a factor of two about its mean, dimmest at the centre and brightest toward the edges and corners, while the bare and specular wrappers occupy a narrow band around unity. The two four-sided layouts (1 and 2) are nearly azimuthally symmetric, whereas in the single-sided layouts (3 and 4) the $TiO_2$ panel shows the one-sided read-out directly, the region opposite the sensors darkening.

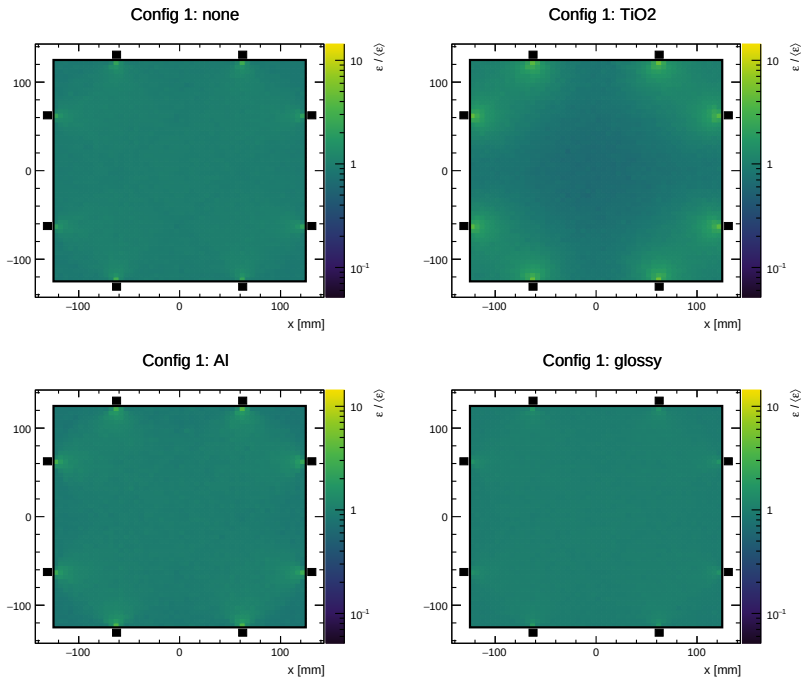



**Figure 19:** Whole-slab efficiency normalised to its plate average, $\varepsilon/\langle\varepsilon\rangle$, for config 1, one panel per reflector (5 mm cells, the same binning as Figs. 14–17, on a shared *logarithmic* colour scale). The slab outline and the SiPM positions (black squares), or for config 4 the WLS fibre clamp (red bar) with its read-out SiPM, are overlaid as in Fig. 17. The $TiO_2$ panel spans well over a factor of two about its mean, dimmest at the centre and brightest at the edges and corners; the bare and specular wrappers occupy a narrow band around unity, confirming that their position dependence is weak.

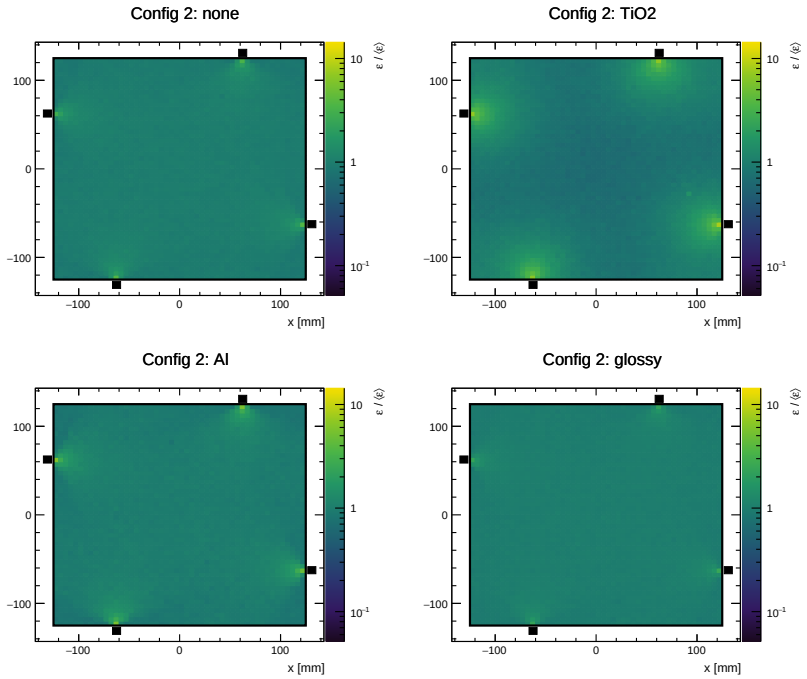



**Figure 20:** As Fig. 19 but for config 2 (four SiPMs, one per lateral face). The patterns are almost indistinguishable from config 1—the four-sided read-out is effectively azimuthally symmetric—with only the overall normalisation (hence the absolute efficiency) reduced.

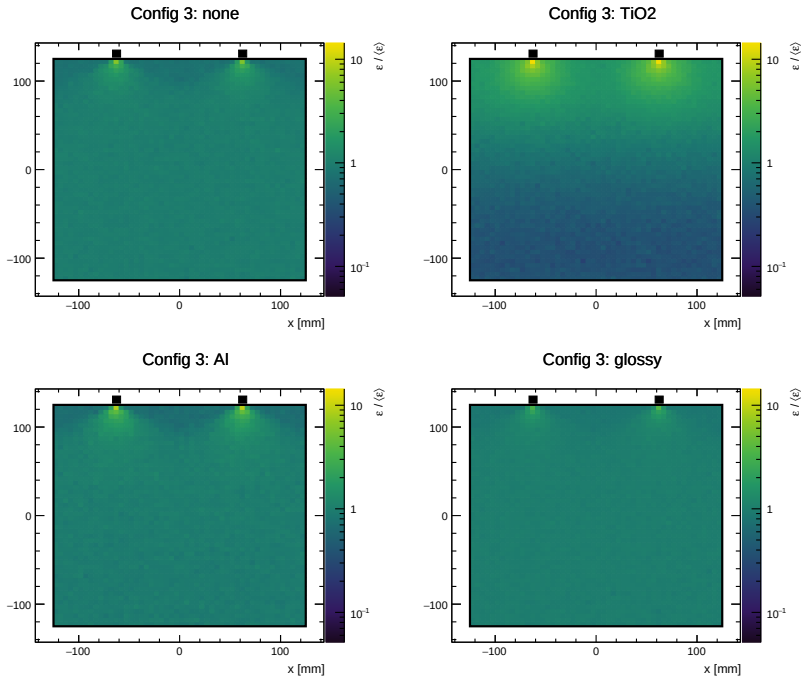



**Figure 21:** As Fig. 19 but for config 3 (two SiPMs, both on the top $+y$ face). The one-sided read-out breaks the azimuthal symmetry: the $TiO_2$ panel shows a pronounced top–bottom gradient, with the response falling toward the $(-y)$ face opposite the sensors. The bare and specular wrappers close most of this gap.

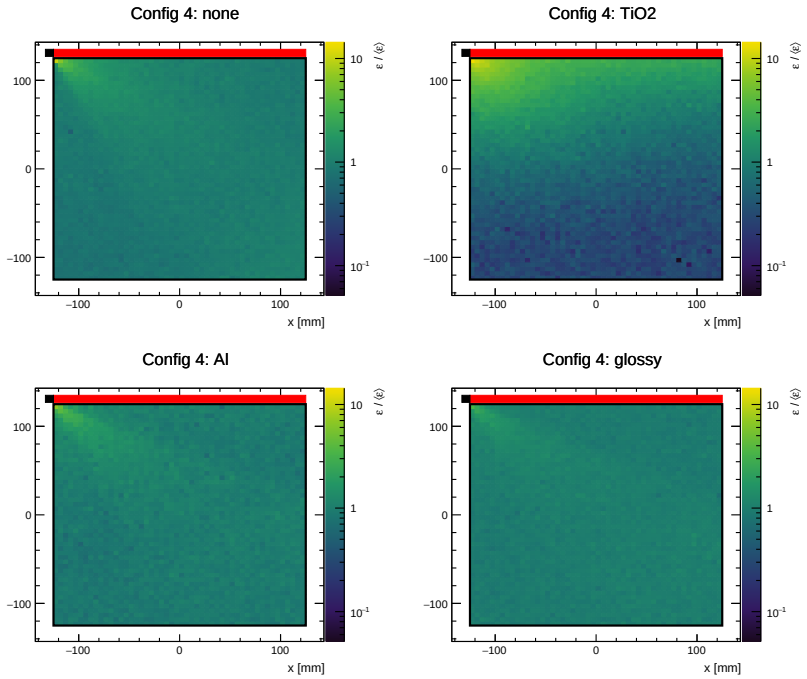



**Figure 22:** As Fig. 19 but for config 4 (single SiPM + WLS fibre clamp along the top $+y$ edge, read out at the top-left corner). The single fibre read-out gives the $TiO_2$ panel the strongest asymmetry of all four layouts, the response falling from the middle of the plate toward the $(-y)$ side opposite the sensors; this is the structure that gives config 4 the largest $\kappa_{2D}$ of the study.

### 4.4 Arrival-time structure and pulse persistence

The collection-efficiency maps above treat every detected photoelectron as a point; the new per-photoelectron timing columns of the `sipmHits` tree let the *arrival-time structure* of the detected light be examined as well. Figure 23 shows the oscilloscope-style persistence map of the arrival-time profile for the four layouts (glossy paper). For each event the per-SiPM arrival times are summed over the layout's sensors and histogrammed in time, giving a trace of photoelectron count versus arrival time; stacking events and recording, for every 0.1 ns time bin, how many events showed a given count, yields a two-dimensional map whose abscissa is the arrival time, whose ordinate is the instantaneous photoelectron count, and whose colour is the number of events. The rising edge and the decaying tail therefore show how the detected light builds up and dies away, and the vertical spread at a given time is the event-to-event fluctuation of the instantaneous rate. For the three directly-read layouts the prompt edge peaks between 4.1 ns and 4.3 ns, reaching instantaneous maxima of 54, 47 and 42 photoelectrons per 0.1 ns bin in configs 1–3; the single-fibre WLS read-out (config 4) peaks markedly later, at 9.6 ns and at only 6 photoelectrons per bin. All four panels share a single linear amplitude axis of 0–30 p.e. at 1 p.e. per bin; the very top of the prompt peak in configs 1–3 lies above this range and is off-scale. About 30% of the light reaches the sensors within 5 ns and 60% within 10 ns for configs 1–3, against 12% and 33% for config 4, while the late ($t > 20$ ns) tail carries 11–15% of the photoelectrons in every layout.

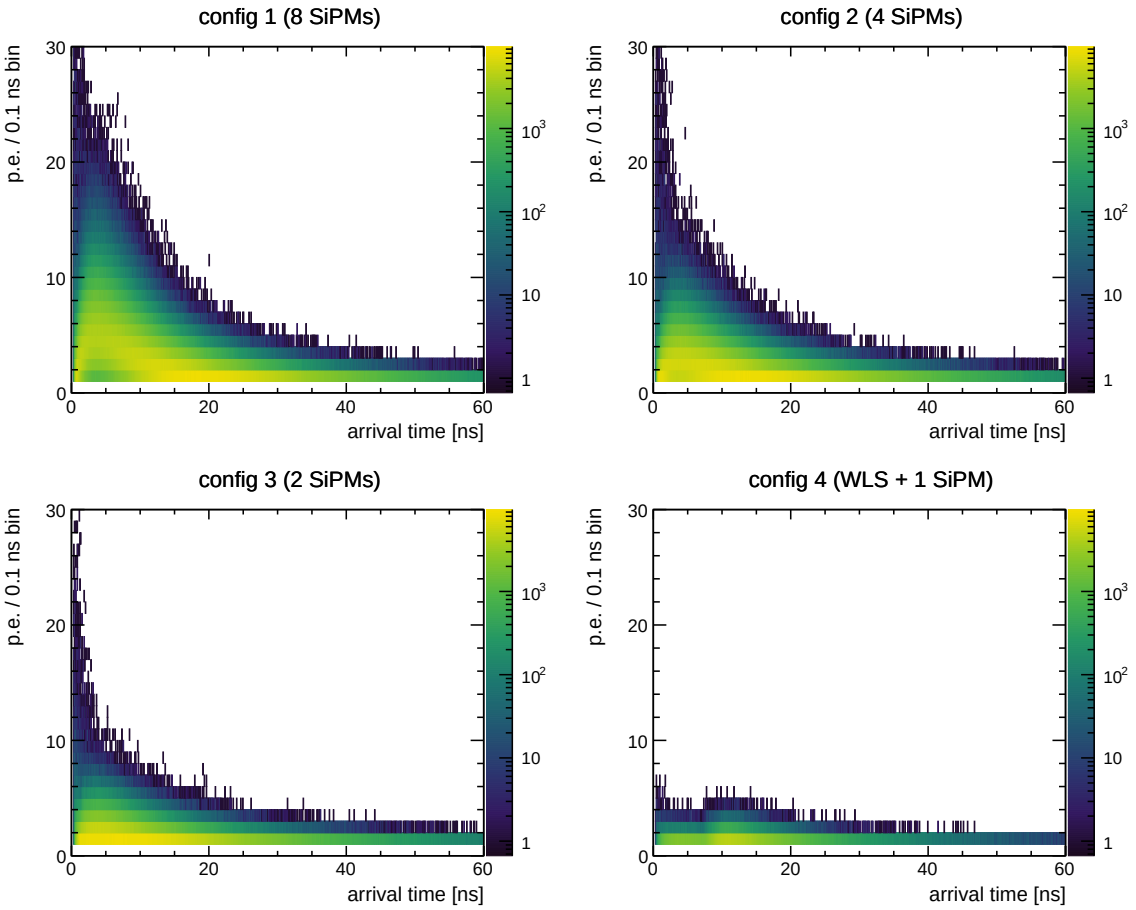



**Figure 23:** Arrival-time persistence map of the detected photoelectrons, one panel per layout (glossy paper, whole-plate irradiation). The abscissa is the global arrival time, the ordinate the number of photoelectrons in the corresponding 0.1 ns time bin (summed over the layout's SiPMs), and the colour the number of events reaching that point (shared logarithmic scale). Empty time bins are omitted; the ordinate is shared across the four panels and linear over 0–30 photoelectrons per bin (1 p.e. per bin), the arrival-time axis over 0–60 ns. The prompt rise reflects the direct scintillation and Čerenkov light, while the slower tail is the multi-bounce transport and delayed (self-absorbed and re-emitted) light. The WLS layout (config 4) builds up roughly 5 ns later than the others, its prompt peak shifted by the fibre absorption and re-emission.

Figure 24 combines the two offline observables: the arrival time and the wavelength of each detected photoelectron. Configurations 1–3 collect the blue scintillation spectrum directly, with a photoelectron-weighted mean of 459.5–460.0 nm, whereas the Y-11 WLS read-out of config 4 is green-shifted to 486.2 nm, consistent with the steady-state spectra of Fig. 8. In every layout the spectrum reddens with arrival time: the mean wavelength rises from 454 nm ($t < 5$ ns) to 470 nm ($t > 20$ ns) for configs 1–3, and from 462 nm to 492 nm for config 4, while the rms spread narrows (38.8 → 35.6 nm and 37.0 → 33.0 nm, respectively). The depleted blue edge of the late light is the signature of self-absorption along the multi-bounce path and, in config 4, of the additional wavelength-dependent attenuation of the WLS fibres.

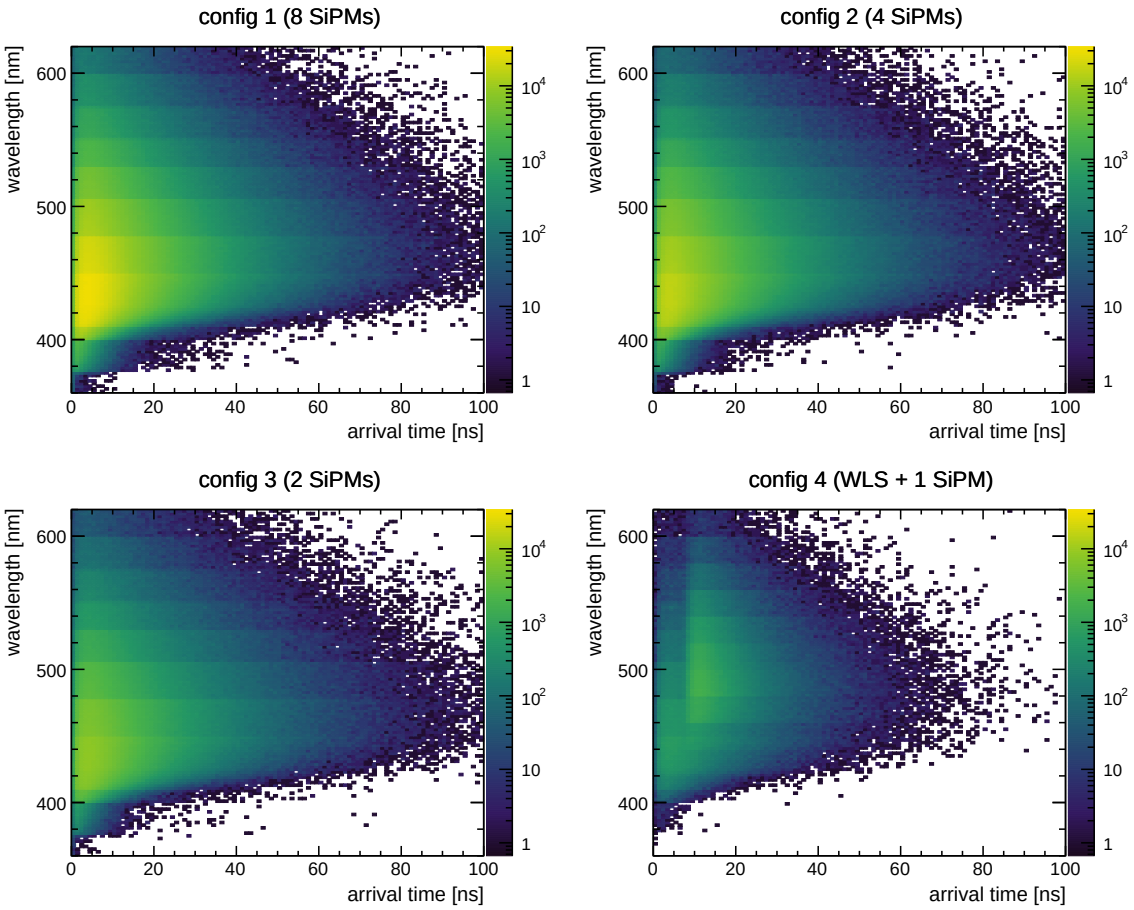



**Figure 24:** Detected wavelength versus arrival time, one panel per layout (glossy paper, whole-plate irradiation), on a shared logarithmic colour scale (number of photoelectrons). Configurations 1–3 populate a narrow band around the 460 nm blue scintillation spectrum; config 4 is green-shifted by the Y-11 WLS fibres. The time dependence of the mean wavelength traces the self-absorption of the shorter-wavelength light over the multi-bounce path (from 454 nm to 470 nm in configs 1–3 and from 462 nm to 492 nm in config 4, between the early and late arrival windows).

## 5 Caveats and open items

**Intermittent multi-threaded stall.** During the matrix production runs a rare stall was observed: on one occasion a single worker thread entered a tight loop (100% CPU) while the remaining threads waited at the event-level barrier, freezing the run. The condition is intermittent—the same macro completed normally on a clean re-run—and is consistent with a pathological optical track (e.g. a photon bouncing in a near-degenerate configuration) rather than a deterministic defect. Because a single stalled task would consume the entire wall-time allocation on a batch system, the production workflow wraps every task in a timeout with automatic retry. The matrix results reported here are unaffected: all 48 files completed (four layouts × four reflectors × three positions), and the frozen run was discarded and regenerated.

**The bare plate is not the weakest collector.** Modelled with a real 1.2 mm air gap between the polished plate and the black holder, the bare configuration is a near-lossless TIR light guide and is the *second*-brightest wrapper in configs 1–3 (and the brightest in config 4), not the weakest. This disagrees with Ref. [1], whose bare plate gives by far the smallest signal. Reproducing that suppression would require the plate–holder interface to be optically *lossy*, i.e. either a substantial real contact area (frustrated TIR, the opposite of an air gap) or a rough/frosted plate surface. A dedicated loss study indicates that the result is robust against absorption—the plate would have

to be an order of magnitude more absorbing to fall below $TiO_2$—and that erasing TIR would require an effective surface roughness $\sigma \approx 0.15\,\mathrm{rad}$ (a ground, not polished, surface), far larger than the paper's "polished plate" description implies. We therefore regard the air-gap model as the physically defensible reading of the stated construction (a plate simply clamped in its holder) and flag the discrepancy rather than tune the surface to force agreement.

**Position trend in $TiO_2$.** For the air-gap bare plate and the mirror-like wrappers (Al, glossy) the response is essentially flat across the three patches (contrast $<6\,\%$ for the specular wrappers and $<18\,\%$ for bare). The diffuse $TiO_2$ paint is the only wrapper with a clear position dependence, and its sign depends on the layout: in the four-sided layouts 1 and 2 it shows corner > edge > centre (a factor $\approx 1.7$), whereas in the single-sided layouts 3 and 4 it shows centre > edge $\approx$ corner. Since `groundfrontpainted` is purely Lambertian (no TIR, no specular reflection), the sign follows the geometry of the nearest painted walls relative to the illuminated sensors. No corresponding trend is reported in Ref. [1] and it should be confirmed with independent simulation settings before being interpreted physically.

**Scale and normalisation.** The absolute efficiencies are model-dependent: they are fixed by the assumed PDE curve, the grease coupling, and the absorbed-fraction of the tabulated reflectivities. The SiPM breakdown-voltage and gain (which drive the PDE) are also temperature-dependent [9], an effect not modelled here. The robust results are the *ordering* ($TiO_2$ weakest, glossy brightest of the wrappers) and the finding that a polished plate in a non-contacting holder is *not* suppressed.

## 6 Summary

A Geant4 optical simulation of a $250\,\mathrm{mm}\times250\,\mathrm{mm}\times10\,\mathrm{mm}$ polished plastic scintillator read out by side-mounted SiPMs has been developed and applied to a full matrix of four SiPM layouts and four reflector treatments at three beam positions, complemented by a uniform whole-plate irradiation. The optical physics—unified surface model, finish selection, photocathode PDE via the boundary process, and per-event photon accounting—has been verified against the Geant4 source. The results reproduce part of the experimentally observed ordering: glossy paper is the brightest wrapper and $TiO_2$ paint the weakest in every layout, and the specular wrappers are the most position-uniform. They do *not*, however, reproduce the experimental result that the bare plate is by far the weakest collector. Modelled with its true air gap, the polished plate is a near-lossless TIR light guide, second only to glossy paper in configs 1–3 and the brightest wrapper in the WLS layout. A loss study shows that the bare plate would have to be an order of magnitude more absorbing, or its faces ground to $\sigma \approx 0.15\,\mathrm{rad}$, to be suppressed below $TiO_2$. Keeping the three irradiation patches separate—rather than collapsing them to a single beam-axis radius—shows that the bare plate and the reflective wrappers homogenise the response over the illuminated region (position contrast below $18\,\%$ and $6\,\%$, respectively), while only the diffuse $TiO_2$ paint retains a clear position trend. Extending the scan to the whole plate sharpens this picture: the full-plate contrast is largest for $TiO_2$ ($\kappa_{2\mathrm{D}} = 1.9$–$16.8$), reaches only 1.1–2.4 for the bare plate, and stays closest to unity ($\leq 1.5$) for the specular wrappers.

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